The Role of Artificial Intelligence in Enhancing Startup Communication

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The Definition of AI

1.

1.1 The Definition of AI

It was John McCarthy who first coined the term artificial intelligence in 1956. According to John McCarthy, it is the science and engineering of marketing intelligent machines. Artificial intelligence (AI) is typically defined as the ability of machines to perform human-like cognitive tasks. These can include automation of physical processes such as manipulating and moving objects, sensing, perceiving, problem-solving, decision-making and innovation .[1]

Or in other words, the ability of machines to mimic human intelligence and perform tasks that typically require human intelligence, such as learning,problem- solving, decision-making, and natural language understanding. Figure 1 depicts AI technologies including machine learning, natural language processing, robotics, and com- puter vision. Machine learning is a subset of AI that involves training computer algorithms to learn patterns in data and make predictions or decisions based on the data.[2]

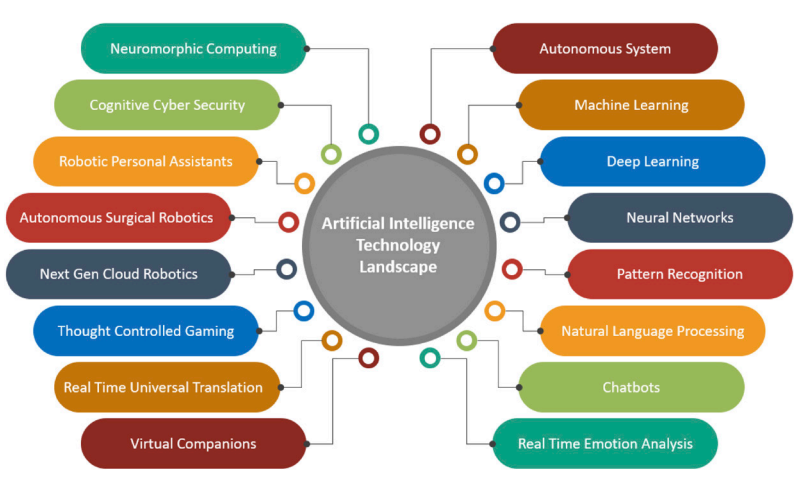


Fig.1 **AI Technology Landscape** Abdulaziz Aldoseri, et al 2023

Besides, AI models rely on large datasets to identify patterns and trends that are difficult to detect using traditional data-analysis methods. This allows them to learn and make predictions based on the data on which they have been trained.

Below are also several AI definitions provided by highly reputable sources:

AI is the theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.[3]

Artificial intelligence (AI) is the intelligence of machines or software, as opposed to the intelligence of living beings, primarily of humans. It is a field of study in computer science that develops and studies intelligent machines. Such machines may be called AIs. [4]

AI is technology that enables computers and machines to simulate human intelligence and problem-solving capabilities. [5]

Artificial intelligence is the ability of a computer or computer-controlled robot to perform tasks that are commonly associated with the intellectual process's characteristic of humans, such as the ability to reason. Although there are as yet no AIs that match full human flexibility over wider domains or in tasks

1. Jie (Kevin) Yan, (2018), Differential Innovativeness Outcomes of User and Employee Participation in an Online User Innovation Community

2. Abdulaziz Aldoseri , (2023), Re-Thinking Data Strategy and Integration for Artificial Intelligence: Concepts, Opportunities, and Challenges

3. Oxford language dictionary (2023)

4. Wikipedia (2021)

5.IBM (2022)

requiring much everyday knowledge, some AIs perform specific tasks as well as humans. [6]

-Artificial Intelligence (AI), a term coined by emeritus Stanford Professor John McCarthy in [1955], was defined by him as “the science and engineering of making intelligent machines”. Much research has shown humans program machines to behave in a clever way, like playing chess, but, today, we emphasize machines that can learn, at least somewhat like human beings do.[7]

6. Britannica (2022)

7. Stanford University (2021)

> **AI is the the science and engineering of making intelligent machines**

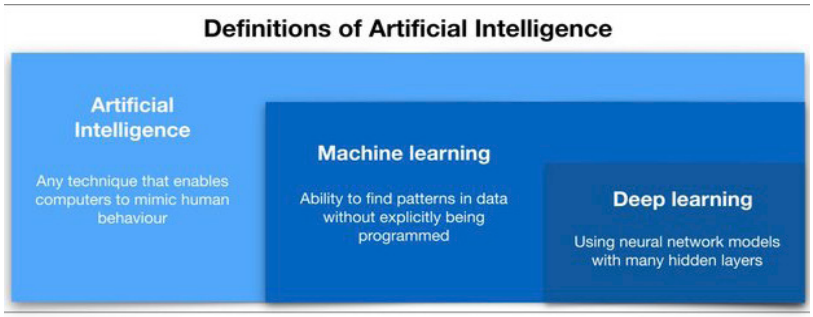


Fig.2
Definitions of AI: AI encompasses both ‘traditional’ machine learning and ‘contemporary’ deep learning concepts
Mangor PedersenMangor Pedersen 2020

Also, Interaction Design Foundation suggests this figure as the definition of AI In 2023, defining that there can be different layers and aspects of AI from deep learning, to Machine learning to finally Artificial intelligence, explaining each.

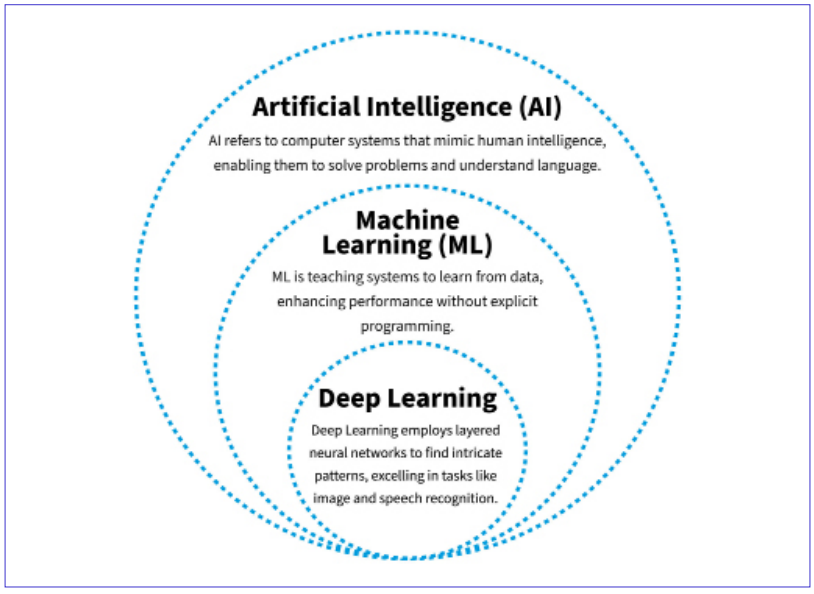


Fig.3
the Definitions of AI
Interaction Design Foundation 2023

Artificial Intelligence (AI) has revolutionized the communication domain, providing advanced tools and solutions that significantly enhance both internal and external interactions within organizations. AI's capabilities in natural language processing (NLP) enable the automation of customer service through chatbots, which can handle a large volume of inquiries efficiently and accurately, thereby improving customer satisfaction and operational efficiency. Additionally, AI-driven analytics can process vast amounts of communication data to provide insights into customer behavior, preferences, and sentiment, allowing for more personalized and effective marketing strategies.

Beside, In the context of startups, the importance of AI in communication cannot be overstated. Startups often operate with limited resources and under high pressure to scale quickly. AI technologies offer scalable solutions that can help these companies manage communications more effectively without the need for extensive human resources. For instance, AI-powered customer relationship management (CRM) systems can streamline sales and customer support processes, enabling startups to maintain high levels of service as they grow. Moreover, AI can facilitate better team collaboration through smart scheduling, task management, and real-time language translation, ensuring that startup teams can work more cohesively and productively, regardless of geographical boundaries.

1.2 The Definition of startup

Term ‘start-up’, which in English has a number of meanings, is also used in Poland to describe a kind of business venture. Word start-up is being used in a variety of contexts, like a new venture in general, company run by a young person or company based on high technology (sometimes specifically in the ICT sector).

In both English and Polish business journals, there are two most known definitions of start- ups. According to Eric Ries it is a ‘human institution designed to deliver a new product or service under conditions of extreme uncertainty’ [8], when for Steve Blank ‘startup is a temporary organization designed to search for a repeatable and scalable business model’ [9]. As we can see, in both cases start-up is considered to be a different type of venture, than just a phase in a company’s development or a business operating in a specific sector.

> **Startup is human institution designed to deliver a new product or service under conditions of extreme uncertainty**

Start-up is understood as a business venture that is either searching for its place on the market or trying to introduce some innovation to it, however there are very few scientific publications concerning the phenomenon of start-ups in Poland. Also quoted definitions haven’t been supported by scientific research, therefore studies presented in this paper are exploratory and their purpose is to fill the theoretical gap by creating a definition of start-up based on research conducted in Poland and compare it to commonly used definitions.

According to a study that have been done by Michał Adamczyk in 2021, start-up is a special kind of organization, which acts on three levels:

1. Operational level: Conducting market experiments and improving business models.

2. Tactical level: Satisfying investors and gaining more funding.

3. Strategic level: Converting innovation into new valuable products and introducing it to the market.

According to that definition, to consider something as a start-up, it must have external financing. Therefore, a couple of people that are trying to build something in a garage can’t be called a start-up, until they convince someone to put his money in their idea.

Including the requirement of having external funding to the start-up definition also has a very practical value. It is possible to explore and study a start-up defined as above using official public records, because in most cases investment is manifested in the creation of a public company which is obliged to submit annual reports that can be easily reviewed.

It seems important to emphasize that although a start-up is based on innovations, it does not necessarily need to create them on its own. Innovation itself can be either made or acquired, but the main purpose of a start-up is to commercialize it, by creating a product that generates real value for customers

8. RIES, E.,(2011), The Lean Startup. New York: Crown Business

9. BLANK, S, (2014), Corporate Acquisitions Of Startups - Why Do They Fail? Forbes

on a global scale. [10]

Beside, The term startup refers to a company in the first stages of operations.[11]

Startups also are young companies founded to develop a unique product or service, bring it to market and make it irresistible and irreplaceable for customers.

Rooted in innovation, a startup aims to remedy deficiencies of existing products or create entirely new categories of goods and services, disrupting entrenched ways of thinking and doing business for entire industries. That's why many startups are known within their respective industries as "disruptors."12

A startup is defined as a small-medium-sized enterprise existing for less than a decade that is privately held and values market innovation and the implementation of unique and scalable business models. [13]

On the other hand, According to the well-known definition by Steve Blank (Blank, [2013]), a startup is a company, a partnership or a temporary organization designed to search for a repeatable and scalable business model. [14]

Startups are possibly the most exciting type of business because they provide a unique opening for global influence and growth. Many individuals could only dream of this possibility a few decades ago. Artificial Intelligence (AI) automates jobs and enables new services worldwide through low-cost computer power, cloud computing services, big data expansion, and breakthroughs in machine learning (ML) and associated processes. AI can alter and enhance the way governments, corporations significantly, and individuals offer services, collect information, and plan and operate. [15]

In the startup environment, a business that's still in the idea phase can potentially have a tremendous impact in the upcoming five years – will have a turnover of a billion dollars. There is a lot of unnecessary propaganda around startups. People who have worked in the startup scene for a long time often suffer from such propaganda. They are doing what they are supposed to do because they believe in future gains. [15]

10. Michał Adamczyk, (2022), An attempt to define the concept of start-up company based on inductive research

11. MITCHELL GRANT, (2024), What a Startup Is and What's Involved in Getting One Off the Ground

12. Rebecca Baldridge, (2022), What Is A Startup? The Ultimate Guide

13. Linjuan Rita Men, et al, (2021), Engaging Startup Employees via Charismatic Leadership Communication: The Importance of Communicating "Vision, Passion, and Care"

14. Irene Di Bernardo, et al, (2021), AI AS A BOOST FOR STARTUPS COMPANIES: EVIDENCE FROM ITALY

15. Akshay Swami, et al, (2022), Impact of Artificial Intelligence on Next-generation Startups

1.3 Communication in startups

This study endeavors to examine the role of Artificial Intelligence in facilitating communication within startup enterprises. To provide context, it is imperative to delineate the concept of communication within the context of startup operations first.

A communication strategy is a plan that describes how an organization or an individual will communicate with its audience. This includes a set of goals and objectives, key messages, communication channels, tactics, and metrics to measure the success of your communication efforts.

A communication strategy is designed to help an organization achieve its communication objectives, such as informing, educating, persuading, and engaging target audiences. It can be used for various purposes, including promoting new products and services, creating brand awareness, crisis management, and influencing public opinion.

Effective communication strategies are tailored to the specific needs and preferences of the target audience and take into account the broader social, cultural, and political context in which they operate. We also rely on various communication channels such as social networks, email, traditional media, events, and advertising to reach our target audience. [16]

Lulu Cheng Meservey, VP of Communications at Substack and former co-founder of TrailRunner International says that, For a startup, the purpose of Communications is to help make money. Even if your company has an altruistic mission, which it hopefully does, you're not going to have the impact you want unless you can make money, grow the business, and thrive as an enterprise. Having an intentional comms strategy is about making sure that your communications are aligned with this goal.he adds, Communications should be a priority before your startup has even launched. That's when you can start positioning yourself as the founder of your future startup.[17]

> **"For a startup, the purpose of Communications is to help make money. Even if your company has an altruistic mission, which it hopefully does, you're not going to have the impact you want unless you can make money, grow the business, and thrive as an enterprise"**

16. The BenTech, (2023), An Overview of Communication Strategies

17. Lulu Cheng Meservey, et al, (2024), Comms Strategy and Tactics for Startups

1.4 Statement of the Problem

In recent years, the rapid advancement of Artificial Intelligence (AI) technologies has revolutionized various aspects of business operations, offering unprecedented opportunities for innovation and efficiency. Particularly within the realm of startups, where effective communication plays a pivotal role in success, there is a growing interest in leveraging AI to enhance communication strategies. However, despite the increasing adoption of AI tools by startups, there remains a gap in understanding the specific role and impact of AI on startups' communication processes.

This study aims to address this gap by investigating the role of Artificial Intelligence in enhancing startups' communication. Specifically, the research will delve into how AI technologies such as natural language processing (NLP), chatbots, and machine learning algorithms are utilized by startups to optimize internal and external communication channels. Moreover, the study seeks to explore the potential benefits, challenges, and implications associated with the integration of AI into startups' communication strategies.

Key questions:

How are AI technologies currently being employed by startups to enhance communication with internal stakeholders (e.g., team members, management) and external stakeholders (e.g., customers, investors)?

What are the perceived benefits and also challenges and barriers of integrating AI into startups' communication processes, and how do these benefits contribute to the overall success and growth of startups?

By addressing these questions, this research endeavors to provide valuable insights into the evolving landscape of startups' communication in the AI era. Furthermore, the findings of this study aim to inform startups, entrepreneurs, policymakers, and researchers about the opportunities and challenges associated with harnessing AI for communication enhancement, ultimately facilitating informed decision-making and fostering innovation in the startup ecosystem.

State of The Art

2

2.1 The Growing Influence of AI in different stages of startup establishments

In literature, very little attention has been paid to the study of the connection between startup companies and artificial intelligence while noting also its relationship with communication. However, there are several contributions, such as the case studies analyzed, which increase the thesis that Artificial Intelligence is a key, useful and successful engine for an innovative startup company in its different stages of establishment.

The flurry of interest in AI is triggering a variety of reactions, everything from excitement about how the capacities may quickly increase the startup businesses [18], making startups more and more scale ups.[19]

Artificial Intelligence is rapidly being employed in startups to build cost-saving systems and procedures, saving them money while gaining a competitive advantage. AI is being utilized in startups to address unique difficulties while launching a business. This involves developing a solid client base, recognizing the vast quantity of data accessible, and exceeding sales and productivity targets. Startups today have access to more sophisticated tools and software to assist them in achieving consistent and sustainable development through the use of AI technologies.[20]

At the startup level, some of the advantages of AI include the rapid discovery of patterns in large amounts of data, rapid visualization and analytics, improved product design, the delivery of precise insights, and many others. These advantages are predicted to bring about new service levels, increased profit, business expansion, improved efficiency, and cost structures.[21]

Innovation, knowledge, and entrepreneurship are the three elements that drive Economics deals with dynamic processes causing qualitative transformation. Bernardo D. et al. explained how artificial intelligence might operate as a catalyst and engine of change in markets through the formation of startups. The characteristics of artificial intelligence align with the requirements of new initiatives. In this study, a large number of Italian startups are profiled, as well as a comparison of startups in the same industry, with a focus on those who use AI and those that do not, in order to identify differences in economic performance, customer perceptions, and business model quality.

Many startups are actively utilizing artificial Intelligence (AI) to transform the manufacturing process, provide better and more innovative services to clients, and change the way businesses function. The traditional link between manufacturing facilities and human resources is shifting as products and services have become totally digital. AI will become product components such as capital and labor and enhance productivity. The new technology will also significantly influence startups' competitive standing. Investing in AI cuts expenses and boosts the competitive edge of innovative enterprises. AI is now undergoing substantial changes in terms of innovation. Instead of coming from global corporations in the past, innovation today comes in big numbers from research institutes, internet platforms, and startups.[22]

History shows us that the main technology companies such as Apple, Google and Amazon that are highlighting artificial intelligence today at the time of their birth were startups.

18. Agrawal et al, (2017) What to Expect from Artificial Intelligence Technology

19. Rinzwana.M,(2019), From Startup to Scale Up: The Key Challenges

20. Oswald. G, et al, (2017), Shaping the Digital Enterprise

21. Irene Di Bernardo, et al, (2021), AI as a Boost for Startups Companies: Evidence From Italy

22. Enholm, E. et al, (2021) "Artificial Intelligence and Business Value

Since the launch of the first products on the market, these already showed a predisposition for future uses of artificial intelligence systems. An example can be the use of data analytics and algorithms is another critical application of artificial intelligence.[23] These algorithms predicting the products that specific clients may buy next can be used to optimize cross-selling and up-selling. For example, Netflix achieved impressive results from the algorithm it uses to personalize recommendations to its [100] million subscribers worldwide. Netflix estimated these customized search results saved it $[1] billion in annual revenues that would otherwise be lost to canceled subscriptions.[24] On the other hand, Companies such as Amazon Prime, Netflix, and Hotstar would not exist without their AI- powered proposal engines. Startups seek new AI opportunities in areas like product forecasting, digital shop setups, supply chain management, and consumer purchasing patterns, among others.

Sonia N. et al. presented the survey of the top [200] AI startups demonstrating the impact of advanced AI research and innovation on the worldwide market. According to the report, the AI wave has begun, and the hunger for AI development is growing exponentially. AI investment has been steadily increasing over the previous six years and is expected to continue in the future. [25]

Palanivelu observed the qualitative research approach was utilized by the researcher, who conducted semi-structured interviews with [10] different entrepreneurs from ten different enterprises. The study's primary findings revealed that competitive pressure, media attention, digital maturity, and consumers were the most influential variables in incorporating AI in business as can be seen in Figure [5]. Respondents received different comments on the findings relating to the benefits of incorporating AI in marketing.[26]

23. Pascal; Barkin, Ian; Wirtz, Jochen (2020). "Intelligent automation: Learn how to harness Artificial Intelligence to boost business & make our world more human".

24. McKinsey Global Institute, 2017. "Artificial intelligence: The next digital frontier?"

25. Agrawal A., Gans, J. S., Goldfarb, A. (2017), What to Expect From Artificial Intelligence. MIT Sloan Management Review.

26. Worakamol Wisetsri, (2021), Rise of Artificial Intelligence in Healthcare Startups in India

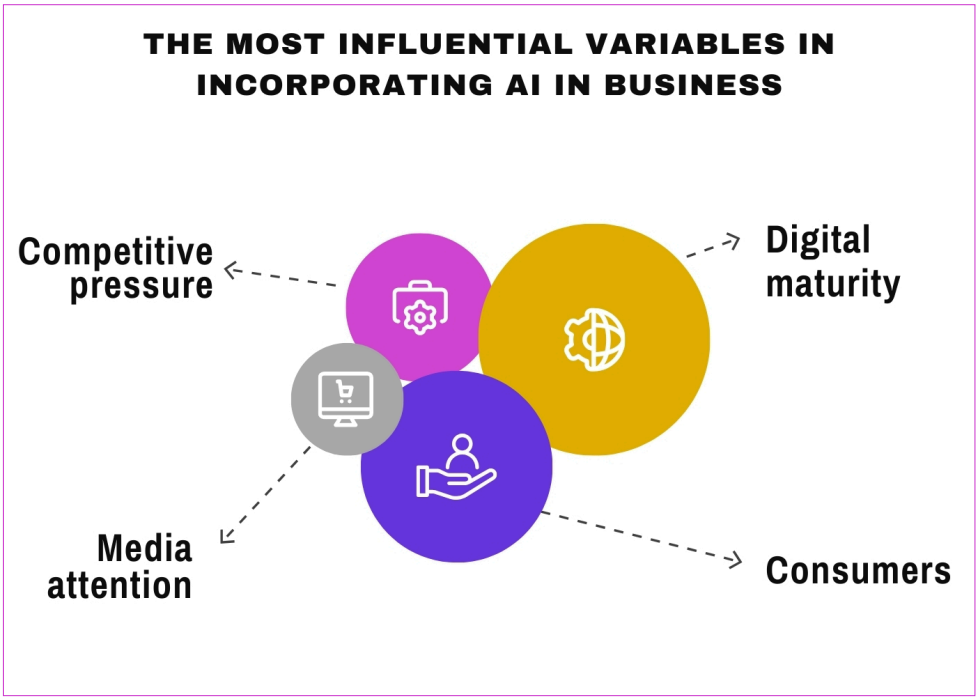


Fig.5 **The most influential variables in incorporating AI in business** W. Wisetsri et al, 2021, Graph created by the auther 2024

In the context of the startup ecosystem, AI assists in developing sales and marketing strategies that result in significant improvements in corporate performance. The above studies emphasize the relevance of AI in business. Furthermore, significant hurdles, ethical considerations, and applications guided businesses looking to implement AI in their operations. Firms should consider the elements and obstacles while incorporating AI into their operations.Wael Basri discussed the influence of artificial intelligence-assisted social media marketing on the performance of small and medium-sized businesses in Saudi Arabia. They used a survey approach to gather, analyze, and interpret primary and secondary data. Participants were Saudi Arabian company owners and workers from startups and small and medium-sized firms .[27]

Partial least squares structural equation modeling was used to examine the data and the findings were that artificial intelligence-assisted social media marketing is becoming more popular among Saudi startups and small and medium enterprises, accounting for an overall increase in the number of customers and customer bases and an additional tertiary effect of increased profitability. Artificial intelligence-assisted social media marketing improves business management and the performance of small and medium-sized businesses.[28]

Ida Merete et al. provide a narrative assessment of how firms might use AI and what value-generating methods such AI employs. The end result of this study is divided into three sections. The study believes that, First, numerous AI enablers and obstacles are discovered. These AI adoption antecedents include technological, organizational, and environmental resources and situations. Second, several AI use cases are distinguished. AI technology may be used by businesses to automate processes or enhance humans for internal or external objectives. Internal purposes refer to the use of AI to enhance internal company operations in situations where the client does not have direct touch with the AI-solution. Furthermore, employing AI for external purposes entails applying it in products and services that directly touch clients.[29]

Paliwal M et al. anticipated artificial intelligence will be able to perform everything a person can, but with more accuracy, and this will be possible via the use of cutting-edge futuristic technology. He claims that Artificial intelligence in corporate operations enables organizations to handle their business prospects swiftly, eliminates mistakes, promotes transparency, and raises revenues by leaps and bounds. It's tough to predict where this technology will generate new jobs in the future .[30]

27. V. Palanivelu et al, (2020), Artificial intelligence (AI) applications for marketing: A literature-based study

28. W. Basri et al, (2020), Examining the Impact of Artificial Intelligence (AI)

29.N. Anute et al, (2021), Artificial Intelligence and Business Value: a Literature Review

30. N. Anute, M. Paliwal, M. et al, (2021) "Impact of artificial intelligence and machine learning on business operations

"artificial intelligence will be able to perform everything a person can, but with more accuracy, and this will be possible via the use of cutting-edge futuristic technology. Artificial intelligence in corporate operations enables organizations to handle their business prospects swiftly, eliminates mistakes, promotes transparency, and raises revenues by leaps and bounds."

Nishad Nawaz describes how artificial intelligence and robotics technologies can fully automate business processes such as finance, human resources, manufacturing, and so on, and how automation can reduce labor and related costs. On the other hand, it will increase the opportunity for startups to have a massive amount of data and information to extract many insights of knowledge by using collected data and will be helpful in data mining and machine learning algorithms.[31]

On the other hand, the essay that has been published by fastercapital tells that AI and machine learning can help startups overcome these challenges and improve financial management. Financial Forecasting, Expense Managemen-t,Fraud Detection and Risk Management and Automated Invoicing and Payments as can be seen in Figure 6.

By leveraging AI and machine learning in financial management, startups can optimize resource allocation, improve cash flow management, and mitigate financial risks.

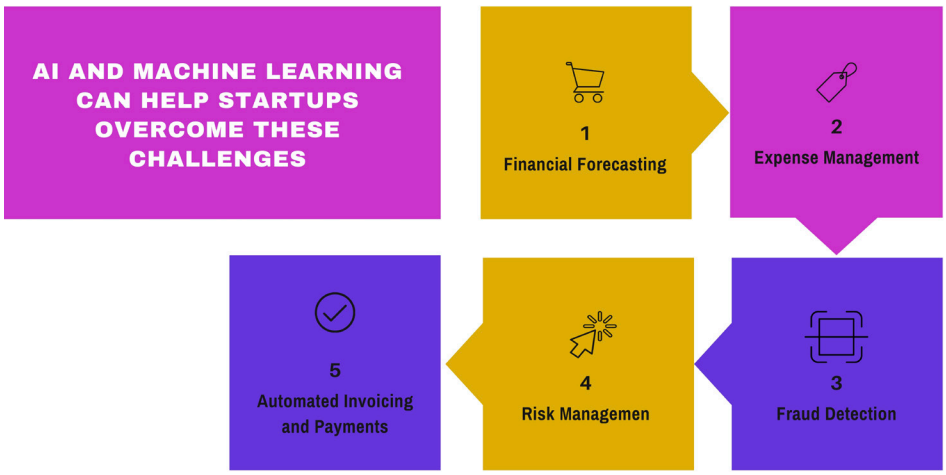


Fig.6
AI and Machine Learning in Startups
https://fastercapital.com
Designed by the author
2024

31.P. Dey, (2022), Impact of AI-focussed technologies on social and technical competencies for HR managers

32.Nageswara Rao, G, et al, (2021)

33. Ali & Anwar, (2021), Business Communication Strategies: Analysis of Internal Communication Processes

Finally, the overview of artificial intelligence as a driver of innovation and success for startups may be enhanced by considering user evaluations and their customer experience, contentment, and loyalty, both in the B²B and B²C markets.[32]

2.2
Role of (strategic) Communication in Startups

Communication is defined as exchange of ideas, information and knowledge between two or more individuals to arrive at a mutual understanding [33], also an information exchange by symbols, signals or words. It is the interchange of viewpoints, thought, belief and facts regarding the common purpose, efforts and interests. [34] Communication may be explained as any behavior that results in an exchange of meaning .[35] Or the practice to transmit information between employees and organizations so based on the meaning of the message the responder can reply. [36] or the action of passing particular information from one individual to another.[37]

Communication also is defined by many authors as the exchange of information between recipient and sender by which a message is directed from a point to another point and the communicators are associated all together through communication channels. Internal communication is defined by Rahman, [38] as necessary management activities in all organizations as it is considered as a vital factor for all employees at all levels in organizations to obtain information regarding their tasks and duties.[33]

Internal communication defined , as an outline of meanings and the individual interaction that take place within organizations [39]. As can be seen in Fig [7].

34. Khan & Abdullah, (2019), The impact of staff training and development on teachers' productivity

35. Anwar & Abdullah, (2021), The impact of Human resource management practice on Organizational performance.International journal of Engineering, Business and Management,

36. Prabhu et al. (2020), Segmenting the manufacturing industries and measuring the performance: using interval-valued triangular fuzzy TOPSISmethod

37. Ganeshkumar et al. (2019), Business Analytics and Supply Chain Performance

38. Abdullah, N. N., & Abdul Rahman, M. (2015). Chinese, Economic Activities and Interests in Developing Countries

> **"Communication is the exchange of information between recipient and sender by which a message is directed from a point to another point and the communicators are associated all together through communication channels."**

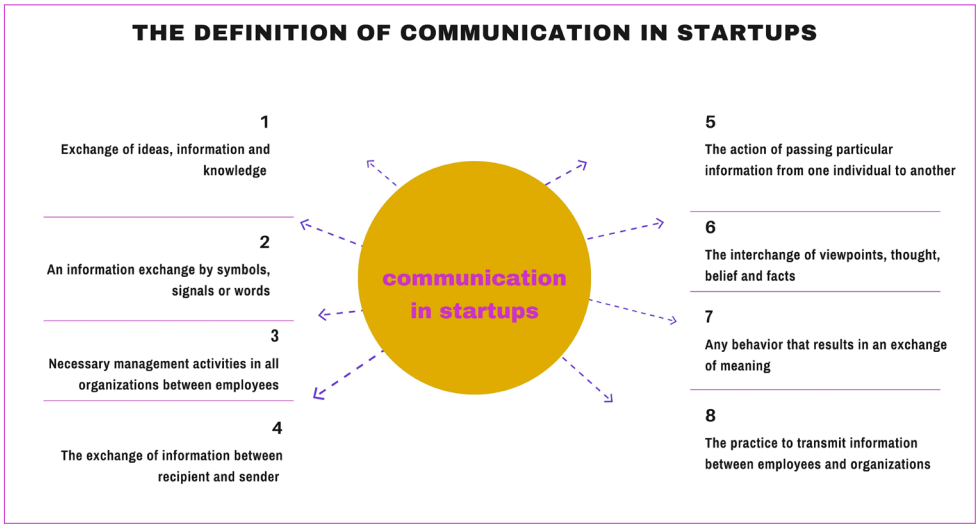


Fig.7 **The definition of communication in startups** Graph created by the auther 2024

Communication is considered to be the base of a startup's operation. The process of communication will have a huge role in maintaining the organization.[40]

Communication within an organization is such an essential aspect of human carrying out that it is almost impossible to imagine a workplace without it. It is through interaction and communication with managers and other employees that a workplace can be maintained. Being an effective communicator is known as a necessary asset in getting along with individuals and obtaining a successful occupation. [41] In the business world, effective communication is seen as an important element of a successful organization. In order to have effective communication within an organization, both types of communication should be included; internal communication such as coordination between departments, among employees and between manager and employees, and external communication such as suppliers, shareholders, agencies and customers. In terms of poor communication, the cost will be very high because it will cause negative results that decrease organizational performance. [35] As it is explained in fig 8.

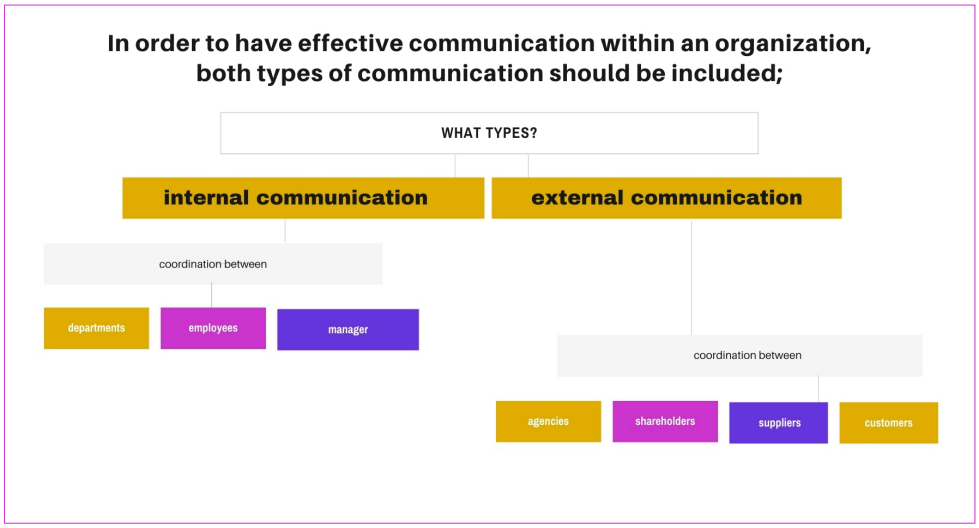


Fig.8 **Different types of effective communication within an organization** Graph created by the auther 2024

The argument that communication is not as important in the early stage of startups as the focus is often on financial and production issues has been a common misconception among many startups. However, an increasing number of entrepreneurs today, especially among the younger generations, have increasingly recognized the critical role that communication plays in startup identity and brand building, customer acquisition, culture cultivation, and stakeholder relationship management, among others. As an entrepreneur remarked, "If you don't think about communication from the beginning, you have to put in the effort several times more later."

> **"If you don't think about communication from the beginning you have to put in the effort several times more later."**

39.Saleh, P. et al, (2021). Factors affecting the Success of Female Entrepreneurs in Kurdistan.

40. Abdullah & Othman, (2019), Impact of the foreign direct investment on the economy of the United Kingdom

41. Top & Ali, (2021), Customer satisfaction in onlin meeting platforms: Impact of efficiency, fulfillment, system availability, and privacy

42. Linjuan Rita, et al, (2019), Best practices of startup strategic communication

Many entrepreneurs lack a comprehensive understanding of what startup public relations or strategic communication entail unless their business is in the communication industry. Thus, they need to be educated regarding what communication really is and, more importantly, what their particular communication needs are. They will likely make startup communication priority high in the company only when entrepreneurs truly recognize what communication can do for their business. Previous literature has revealed the key functions of startup public relations and strategic communication. The functions vary from establishing the startup identity, brand, and culture, building startup image and reputation, generating word of mouth, managing stakeholder relationships, identifying issues, and managing crises, to more micro-level roles of customer acquisition, fostering, and retention and employee communication, among others. As Health (2005) noted, depending on the challenges startups face in the early stage, they can strategically prioritize some of their communication functions by investing considerably more time, budget, and personnel, [42] as described in fig 9.

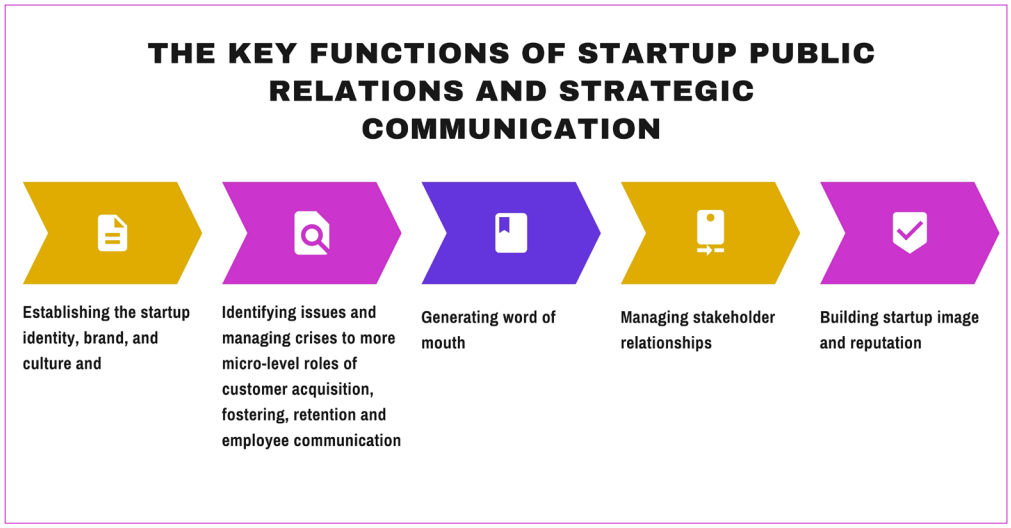


Fig.9
The key functions of startup public relations/ strategic communication
Linjuan R et al, 2019, Graph created by the auther
2024

"The functions vary from establishing the startup identity, brand, and culture, building startup image and reputation, generating word of mouth, managing stakeholder relationships, identifying issues, and managing crises, to more micro-level roles of customer acquisition, fostering, and retention and employee communication, among others"

- Strategic communication

Strategic communication is often used to denote an array of communication activities, such as public relations, marketing communication, advertising, and managerial communication. It emphasizes on how an organization can advance its mission through the strategic application of intentional communication.[43]

The term strategic now has afforded new meanings. For instance, in strategic communication, it underscores the communication function that takes on more responsibilities in making managerial decisions, as opposed to a focus on the technical roles. Strategic communication now is deemed as an interdisciplinary concept that advances an organization's mission and its goal/objective achievement through purposeful communication. In the philosophy of organizational strategic communication, stakeholders are expected to be engaged, and sustain collaborative and mutually beneficial relationships with the organization, [28]Specifically, in the context of startups.[43]

According to an analysis and study that has been conducted by Jhoana Córdova Camacho et al, in [2022] on Strategic communication for startups, digital social networks are synonymous with communication. For startups, digital social networks, in addition to being a source of entertainment and information, are a business communication tool. According to the variables identified, they are communication and relationship tools. The use of social networks by startups can be summarized as follows: Interact with customers, as a platform for conducting business as a means of disclosure of companies and their products; [2]. Enable the basis for planning their actions that through the information generated and provided allows developing or redesigning their products, services and promotional content; [3]. Reduce costs of costly disclosure, but rather direct and more far-reaching; and [4]. Exponentially expand its image.[44]

> **"In the philosophy of organizational strategic communication, stakeholders are expected to be engaged, and sustain collaborative and mutually beneficial relationships with the organization, Specifically, in the context of startups"**

There are two reasons why [45], believe knowledge on startup strategic communication should be advanced. First and foremost, strategic communication is crucial for the sustainable growth of a startup business. Strategic communication cannot be separated from any stage of a new venture development. No matter whether the goal is to establish a vision, build a startup brand, or launch and promote products, a startup needs to manage relationships with various stakeholders (e.g., investors, customers, employees, government, media, and the community) and handle issues and crises. Second, although public relations and strategic communication have evolved as disciplines and practices in

43. Kirk HallahanKirk Hallahan et al, (2007), Defining Strategic Communication

44. Jhoana Córdova Camacho, (2022), Strategic communication for startups: analysis of its intervention in the use of social networks

the past decades, most of the theories and knowledge accumulated are based on research conducted in large corporations. Different from established companies, startups bear many distinctions pertaining to its early developmental stage in the business cycle, company size, structure, and lack of identity and resources. The unique characteristics of startups as compared with large and established corporations require specific guidelines for their strategic communication activities. Obviously, the functions that public relations or strategic communication often serves at large companies, such as branding, reputation management, corporate social responsibility, and marketing communications, may be different for startups. A public relations strategy that works for a multinational company may not be suitable for a startup. [45]

On the other hand, as Sven Patzer claims, Communication assumes many roles within a startup, each contributing to its overall success. At its core, effective communication sets the tone for the company's culture, fostering an environment of transparency, trust and collaboration. It serves as the lifeblood that enables efficient problem-solving and facilitates sustainable growth. Moreover, when communication flows seamlessly, it enhances teamwork and ensures that the entire startup operates as a cohesive unit.[46]

Effective communication is the glue that holds the team together. Promoting a shared understanding and aligning everyone towards a common goal, exponentially enhances teamwork and collaborative efforts. A startup operating with a strong foundation of clear and effective communication becomes more productive, efficient and capable of weathering its inevitable challenges. Furthermore, when a startup can clearly articulate its vision, values and goals to investors and clients, it builds trust and credibility, fostering stronger relationships and solidifying its position in the market. [46]

“Communication assumes many roles within a startup, each contributing to its overall success. At its core, effective communication sets the tone for the company's culture, fostering an environment of transparency, trust and collaboration”

45. Linjuan Rita Men, et al, (2019) ,Why startup strategic communication, https://www.researchgate

46. SVEN PATZER, (2023), Why Effective Communication is the Key to Success for Startups

2.3 Overview of Artificial Intelligence in Communication

Communication is the basic process through which people form perceptions of others [47], build and maintain social relationships , and achieve cooperative outcomes, [48] Generative AI that draws from Large Language Models (LLMs) is poised to fundamentally change how we communicate. AI applications like ChatGPT are increasingly used to produce any kind of language, from text messages and social media posts to computer programs and speeches. [49]

On the other hand, an experiment that has been done by a group of scientists in [2023], [50] shows that generative AI, including a commercially-deployed AI system, can have a significant impact on how people communicate with both positive and negative consequences. It has been found that people choose to use AI when given the opportunity, and this increases the speed of communication and leads to more emotionally positive language. However, it has also been found that when participants think that their partner is using more algorithmic responses, they perceive them as less cooperative, less affiliative and more dominant.

For more than [70] years, the study of artificial intelligence (AI) and the study of communication have proceeded along separate trajectories. Research regarding AI has focused on how to reproduce aspects of human intelligence, including the ability to communicate, within the machine . [51]

AI technology heralds the beginning of a new data revolution that integrates computer science, machine learning, and information technology. AI is a novel field of study that offers improvements for consciously controlling strategic corporate operations. Although little is known about the phenomenon, there is a lot of literature on AI, particularly in management contexts [51]. However, recent studies indicate that communicators are using AI in a variety of disciplines, including, but not limited to, tourism, strategic marketing, political communication, public relations, relationship marketing, customer relationship management, and healthcare [52]

The field of communication is one of the many areas where AI holds enormous potential. The promise of AI for practitioners encourages an environment that is conducive to co-creation from collective intelligence, according to Macnamara and Zerfass ([2012]). In essence, AI has transformed professionals from spectators to participants and co-creators with the public through the use of technology. [53]

47. Ali & Anwar, (2021), Business Communication Strategies: Analysis of Internal Communication Processes

48. Khan & Abdullah, (2019), The impact of staff training and development on teachers' productivity. Economics, Management and Sustainability,

49. Anwar & Abdullah, (2021), The impact of Human resource management practice on Organizational performance. International journal of Engineering, Business and Management,

50. Prabhu et al. (2020), Segmenting the manufacturing industries and measuring the performance: using interval-valued triangular fuzzy TOPSISmethod

51. Ganeshkumar et al. (2019), Business Analytics and Supply Chain Performance: Partial Least Squares-Structural Equation Modeling

52. Abdullah, N. N., & Abdul Rahman, M. (2015). Chinese, Economic Activities and Interests in Developing Countries.

53. Saleh, P. F., Ali, B. J., Akoi, S., Najmalddin, B., Ali, R. S., & Anwar, G. (2021). Factors affecting the Success of Female Entrepreneurs in Kurdistan.

- Role of Chatbots and virtual assistants in customer communication

Chatbots are software programs that operate automatically and are capable of communicating with people. They have a wide range of potential uses, from providing customer service to gathering information or directing requests. [54]

By using NLP, LLM Chatbots can mimic human conversation, making interactions more natural and seamless. [55] This means that LLM Chatbots can hold conversations with users, understand their requests, and provide appropriate responses that meet their needs, all without human intervention. NLP technology has greatly expanded the capabilities of chatbots, enabling them to understand the nuances of language and provide accurate, useful responses in real-time.

Chatbots have come a long way since their inception in the midth century, and they have become increasingly sophisticated with the development of artificial intelligence Technologies. [56] LLM Chatbots are now capable of holding natural conversations with humans, understanding the nuances of language, and providing personalized and accurate responses [56]. In the realm of social media, LLM Chatbots have become an essential tool for businesses to engage with customers and provide personalized support. As users increasingly turn to social media for customer service, LLM Chatbots provide a convenient and efficient way for businesses to interact with customers and address their needs. LLM Chatbots can also help businesses automate sales, marketing, and customer service on social media platforms, improving business efficiency, productivity, and profitability. With their ability to provide instant assistance and support, LLM Chatbots have revolutionized customer service in social media platforms, leading to improved customer experiences and increased customer communication.[57]

As AI technology continues to advance, LLM Chatbots are likely to become even more advanced, providing more personalized and human-like interactions with users. As NLP, machine learning, and voice recognition technologies continue to advance, LLM Chatbots will be able to handle more complex tasks and engage in more human-like conversations with users . [58] This will further enhance user engagement, satisfaction, and loyalty, as well as improve business-customer interactions.

On the other hand, the significance of AI-driven chatbots and virtual assistants can not be overlooked, in the context of startups' communication. They revolutionize communication by offering instant engagement, simulating human-like conversations, guiding users, and delivering relevant information. Their efficiency ensures prompt responses, freeing up human resources for more complex tasks and enhancing overall communication effectiveness.[59]

Thus, AI-powered tools can streamline and automate various communication processes, such as answering frequently asked questions, scheduling meetings, and providing updates on project statuses. By handling routine

54. Abdullah & Othman, (2019), Impact of the foreign direct investment on the economy of the United Kingdom

55. Top & Ali, (2021), Customer satisfaction in onlin meeting platforms: Impact of efficiency, fulfillment, system availability, and privacy

56. Eun GoS. Shyam SundarS, (20190, Humanizing Chatbots: The effects of visual, identity and conversational cues on humanness perceptions

57. Anbang Xu, et al, (2017), A New Chatbot for Customer Service on Social Media

58. Prissadang Suta, et al, (2020), An Overview of Machine Learning in Chatbots

59. Rachna Shah, (2023), The Transformative Impact of AI in Communication

tasks, chatbots and virtual assistants free up employees' time, allowing them to focus on more strategic and value-added activities. Additionally, these tools can improve accessibility to information and resources within the organization, enabling employees to quickly retrieve relevant data and insights to support decision-making processes. Furthermore, chatbots and virtual assistants can enhance collaboration and knowledge sharing among team members by facilitating real-time communication and providing instant access to relevant expertise and resources. However, their impact on communication effectiveness may depend on factors such as the quality of their natural language processing capabilities, user interface design, and integration with existing communication channels. Additionally, concerns related to data privacy, security, and ethical considerations may arise, necessitating careful implementation and governance strategies to ensure that these tools contribute positively to the overall communication dynamics within startup companies. As explained in figure 10.

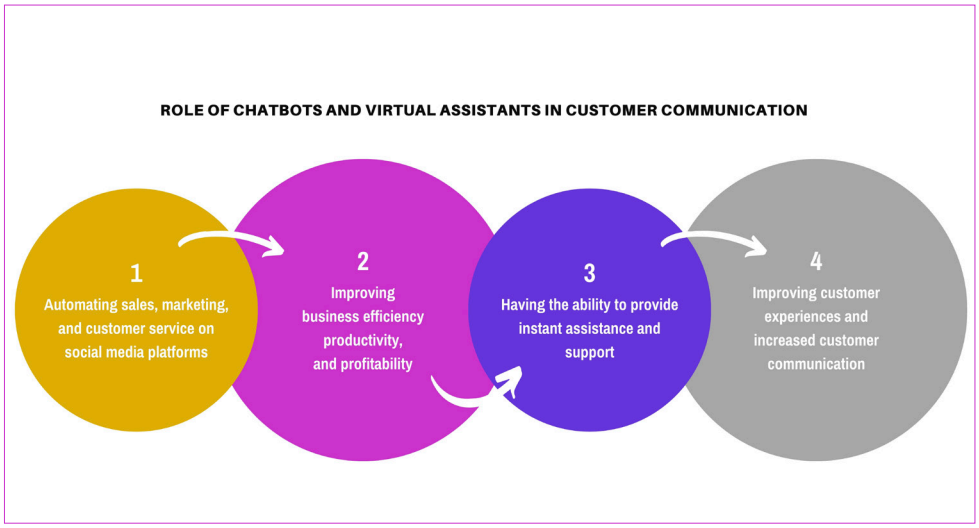


Fig.10
Role of Ai and LLM Chatbots in customer communication
Anbang X et al, 2017, Designed by the author
2024

- Overview of Artificial Intelligence in (strategic) Communication

The strategic communication literature views innovation technology as tools that help organizations advance and strengthen their positions in emerging markets and competition.46 As a result, an organization that adopts innovation is largely responsible for its success. Apart from the other subfields of the communication discipline, the innovative and 'disruptive' integration of augmented technology, or artificial intelligence (AI), in mediated communication has drawn increasing interest from communication scholars (e.g. Brobbey, Ankrah & Kan-

kam, 2021; Moore & Hübscher, 2021; Panda et al., 2019; Zerfass, Hagelstein & Tench, 2021; Zhang et al., 2021). Although some studies 46 have criticized corporate innovation efforts as ineffective, the topic of innovation in strategic communication is still significant and an emerging force.

Over the years, there have been evident advancements and fundamental shifts in how communication is performed. These advancements highlight the dynamic nature of communication as a field and the increasing use of technologies to augment the process of ideation, creation, and transmission of messages in a range of situations, including interpersonal communication, media and journalism, and strategic communication. Innovative technology and the internet are evolving. Based on this, companies have a responsibility to develop and utilize the emergence of innovative technology in their area of business. According to Zerfass and Huck (2007), innovation offers a company the chance to strengthen its in-built capabilities. As a result, innovation becomes a crucial competency for a company because it promotes growth, success, and profitability. [60]

Finally, Artificial Intelligence (AI) has revolutionized the way we communicate. With advancements in Natural Language Processing and Machine Learning, AI has made communication easier, faster, and more efficient. [61]

With the advent of artificial intelligence (AI) and machine learning, we're about to begin a new era for interactions in the work environment. Such rising technologies are creating exciting opportunities, leading in an ecosystem where virtual robots, voice assistants and robotic process automation (RPA) will serve in collaboration with humans to streamline both communications and operations.This will have an enormous impact on both the work we do and the way we communicate with colleagues and clients.

"With the advent of artificial intelligence (AI) and machine learning, we're about to begin a new era for interactions in the work environment"

- different kinds of communication in a startup

As previously mentioned, communication within an organization encompasses various forms and functions. Therefore, this literature review aims to categorize and delineate the diverse types of communication observed within the context of an early-stage startup.

Various types of communication that will be discussed in this chapter include:

Internal communication
Communication with customers
Communication through Marketing
Communication through personalization
Content creation

60. Simone Huck (2007), Innovation, Communication, and Leadership: New Developments in Strategic Communication

61. Eva Rodríguez Vidal, (2024), The Impact of Artificial Intelligence on Internal Communication

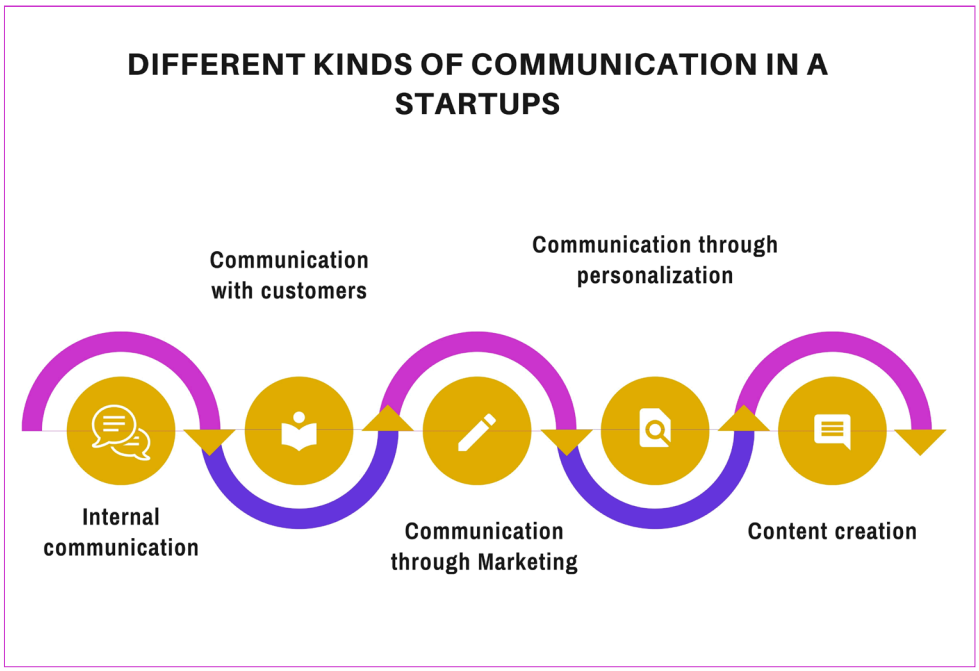


Fig.11
Different kinds of communication within a startup
Designed by the author
2024

- Internal communication

Internal communication is among the fastest growing specializations in public relations and communication management. Its rise began in the 1990s in the US and spread thereafter to Europe growing strength in the new millennium. [62]

In the past, Internal Communication was often seen as a secondary discipline compared to external communication, but this has changed in recent years. Now, most organizations understand the value of effective Internal Communication and recognize that investing in it is investing in their human capital, demonstrating that employees are important and worth listening to, fostering transparent and close bidirectional communication.

Besides, Artificial Intelligence has definitively emerged from laboratories and is increasingly accessible to all companies and individuals. AI is becoming an intelligent revolution, enabling companies to significantly improve their operations through personalized, efficient, and accurate solutions.

Regarding Internal Communication, AI offers a wide range of tools ranging from faster addressing of employee needs, for example, through assistance chatbots, to detecting potential employee satisfaction issues through sentiment analysis. This way, employees can communicate with the company more effectively and receive relevant information quickly and accurately, thus reducing manual workload and improving efficiency in Internal Communication. 49

Specifically, Some companies use AI-based sentiment analysis tools to measure the tone and emotion of employee comments. This allows them to identify areas for improvement in internal communication and address issues proactively.

Some companies also employ AI to personalize Internal Communication based on individual employee needs and preferences. This can improve the employee experience and enhance communication effectiveness through the

62. Ana Tkalac Verčič, et al, (2012), Internal communication: Definition, parameters, and the future

startup ecosystem.

Finally, AI is used to automate tasks that previously required human intervention, such as email sorting and identifying relevant topics in conversations. This can reduce the workload for employees and allow them to focus on more strategic tasks.

Talking about the communication with employees, Machine learning and artificial intelligence will combine knowledge, validate patterns, and optimize and forecast trends. Algorithms based on the technology will choose resumes, identify suitable applicants, create superior employee profiles, understand video interviews, and provide information on persons who appear to be on their way to the interview. Several companies utilize AI to find perfect employees, evaluate the resumes of potential employees, send employment letters to them, and develop professional communications. [62]

According to an essay written by OneTeam agency, AI can help boost internal communication in a number of ways. AI can automate repetitive tasks such as scheduling meetings, sending reminders, and responding to common questions, freeing up time for more critical communication.

AI also can help you with content creation for your (internal) comms channels. With minimum input, you can generate written content, like articles, newsletters, and social media posts. Next to writing content, AI can also help you generate images, video, and audio. Some tools for this can be found further in this blog post. Besides, AI-powered tools can help employees collaborate more effectively by providing them with real-time translation, speech-to-text, and text-to-speech capabilities, making it easier for employees to communicate regardless of language barriers.[63]

On the other hand, AI allows companies to recommend customized training itineraries according to the particular needs of employees, individually adapting their training and career development plans. It also allows the implementation of adaptive evaluation systems to measure different key elements in the development and evolution of employees, considering their personal and individual characteristics in terms of knowledge, competencies and skills, commitment and motivation or work experience and studies. This results in increased reliability and validity of the assessment processes whilst also improving the efficiency of employee training and development.[64]

Thus, In the realm of startup companies, the advent of artificial intelligence (AI) stands poised to revolutionize internal communication processes, fundamentally altering the dynamics of organizational interaction and information exchange. AI technologies offer unprecedented capabilities to enhance the efficiency, effectiveness, and accessibility of internal communication channels within startup environments. By leveraging advanced natural language processing (NLP) algorithms and machine learning models, AI-powered communication tools can analyze vast volumes of data, extract valuable insights, and generate tailored recommendations to facilitate seamless collaboration and decision-making among team members.[65]

63. onpassive.com, (2023), Impact of Artificial Intelligence On Communication

64. Sander Kalkman, (2023), 5 AI tools to help you boost your internal communication

65. knowmadmood.com, (2024), How AI is reinventing talent management in organizations

- Communication with customers

Firms that adopt AI in their processes are able to automate routine functions, especially in the area of sales, but the most interesting aspect is that they are able to enhance their sales through customization, personalization, and improved service quality while increasing effectiveness at the same time [66]. This ability extends to the use of AI as an advanced analytics tool to engage in a number of activities like making tailored-made offers to customers based on their specifications, providing virtual agents to converse online with the customers, and proactively suggesting solutions to issues that the customers might face. [67]

AI comes with numerous potentials that open vast doors for marketing practitioners, which is a must for them in the ever-changing sales environment that continually becomes more complex . [68]

Over the last decades, there have been pronounced changes in consumers' purchasing behavior. Due to increased access to information, consumers can now engage in their own research and set preferred purchasing criteria. A study has also found that B²B customers are usually [50–60]% down the purchasing process prior to reaching a sales representative. [69]

.Additionally, there is an increased adoption of online channels throughout the purchasing process, with findings revealing that about [83]% of buyers access digital channels for the purpose of gathering information as they process across the purchasing process . [70]

The majority of those that purchase via online channels are content-driven, find it comfortable to engage with brands via digital channels, and are technically savvy. [71] Therefore, the need for marketers to adapt their practices towards providing more of a digital solution is highlighted by this shift towards online purchasing.

A significant body of studies has looked at the relationship between artificial intelligence and personalization. In the work of Gao and Liu ([2022]), the focus was on artificial intelligence- enabled personalization in interactive marketing, a journey from the customers' perspective. [72]

66. William C. Moncrief, (2027), Are sales as we know it dying or merely transforming?, https://www.tandfonline.com/doi/full/10.1080/08853134.2017.1386110

67. Andrej Miklosik, et al, (2017), Towards the Adoption of Machine Learning-Based Analytical Tools in Digital Marketing,

68. Anzén & Ekberg, 2020, Understanding how automatized personalization with AI can drive value in B2B marketing

69. Adamson et al., 2012, The End of Solution Sales

70. Gartner, 2018, What Sales Should Know About Modern B2B Buyers

71. Vieira et al., 2019, In pursuit of an effective B2B digital marketing strategy in an emerging market

72. Mariapaola Saponaro et al, (2018), Challenges and Opportunities of Artificial Intelligence in the Fashion World

"In the realm of startup companies, the advent of artificial intelligence (AI) stands poised to revolutionize internal communication processes, fundamentally altering the dynamics of organizational interaction and information exchange."

- Communication through Marketing

As the study that was done in 2022 by Chintalapati et al shows, one of the most affected functions of the AI revolution is digital marketing. [73]

Zerfass et al. (2020) conducted a cross-sectional study to examine the adoption, impact, difficulties, and risks of AI in redefining communication management. Their research included literature on this gap as well as perspectives on innovation from communication professionals. Based on expertise in AI, the rate at which it is affecting communication management, implementation obstacles, and perceived risks, their research selected 2,689 European communication practitioners. The study's conclusions showed that communication practitioners lacked both the personal and organizational competencies necessary to manage the responsibilities and risks associated with adopting AI. The adoption of AI and its effects on startups should be the primary focus of communication leaders. Zerfass et al. (2020) predicted that AI would have a significant impact on the field whereas top management support and championing innovation will help the technology be adopted. The study concluded that their initial study will serve as a springboard for further investigation into how the technology is used in organizations that engage in communication. [74]

Start-ups and small businesses typically face marketing challenges that can be related to a lack of resources, such as marketing skills, time, and money. This also applies to the execution of social media communication strategies that, among other things, require a good understanding of the target customer, content writing and designing skills, a balancing of different social media channels and, in particular, sufficient time to execute. As a result, social media communication activities run by start-ups and small firms often do not achieve the desired results, either because the firm does not use social media channels enough or tries to overachieve.[75]

It is important to mention that, the evolution of AI is set to raise more concerns about security, and the ongoing privacy versus personalization debate is set to expand its scope into softer aspects of marketing. The ability of AI to continuously learn and interpret/forecast the customer buying intents and emotions will help make channels and focus future marketing efforts, leading to extreme automa- tion and personalization. While there will be continued, current, and live debates around deep reasoning, smart AI, continuous/catastrophic learning, and many other attributes pertaining to human AI delegation, this field of study is expected to exponentially evolve. Moreover, the sus- tainability frontier of AI in marketing is expected to become deeper and wider, progressively creating research inputs with stronger actionable insights.[73]

Beside, In an era where consumers seek unique and personalized experiences, generic marketing strategies are becoming obsolete. AI steps in as the driving force behind hyper-personalization by utilizing advanced algorithms and machine learning to create tailored messages that resonate on a personal level. Through AI-driven customer segmentation, marketers can craft campaigns based on granular insights into behaviors, demographics, and psychographics, fostering stronger brand-consumer connections.[76]

73. Srikrishna Chintalapati rt al, (2022), Artificial intelligence in marketing:
A systematic literature review

74. Ansgar Zerfass, et al, (2020), Artificial intelligence in communication management: a cross-national study on adoption and knowledge, impact, challenges and risks

75. Whitler KA (2019) The biggest marketing challenge for startups

Abmatic claims that, The rise of artificial intelligence (AI) has revolutionized the world of marketing, making it possible to personalize advertisements and recommendations to an unprecedented degree. By analyzing vast amounts of data on consumer behavior, AI algorithms can predict what you're likely to buy, what you're interested in, and even what you might need before you know it yourself. This has transformed the way companies approach marketing, allowing them to create targeted campaigns that speak directly to each individual customer. But with great power comes great responsibility, and there are concerns about the ethics of using AI for personalized marketing. In this article, we'll explore the role of AI in personalized marketing, looking at both the benefits and the potential pitfalls of this technology.

There are several benefits to using AI in personalized marketing. One of the primary advantages is the ability to process and analyze vast amounts of data quickly and accurately. By analyzing customer data, AI algorithms can identify patterns and make predictions about customer behavior, preferences, and needs. This enables marketers to create personalized experiences for each customer, such as targeted ads, product recommendations, and personalized messaging.

Another benefit of using AI in personalized marketing is the ability to automate and scale marketing efforts. With AI-powered tools, marketers can create customized experiences for millions of customers simultaneously, without the need for manual intervention. This can save time and resources while also improving the customer experience.

AI also enables marketers to optimize marketing campaigns in real-time. By analyzing customer data in real-time, AI algorithms can make on-the-fly adjustments to campaigns to improve their effectiveness. For example, if a campaign is not performing well, AI can identify the problem and make recommendations for improvement. [76]

Thus, AI-powered data analytics tools can analyze large volumes of consumer data to identify trends, preferences, and behavior patterns, enabling startups to develop targeted marketing campaigns and personalized messaging strategies. By leveraging machine learning algorithms, AI can segment audiences more effectively and optimize marketing content for different demographic groups, thereby increasing the relevance and impact of marketing communications. Additionally, AI-driven chatbots and virtual assistants can engage with customers in real-time, answering inquiries, providing product recommendations, and facilitating transactions, thus enhancing customer engagement and satisfaction. Moreover, AI can automate various marketing tasks such as email marketing, social media management, and content creation, allowing startups to streamline their marketing processes and allocate resources more efficiently. Overall, AI empowers startups to deliver more effective, personalized, and data-driven communication strategies, enabling them to achieve their marketing objectives more efficiently in today's competitive landscape.

76. Jimit Mehta, (2023), The role of artificial intelligence in personalized marketing

> **AI also enables marketers to optimize marketing campaigns in real-time. By analyzing customer data in real-time, AI algorithms can make on-the-fly adjustments to campaigns to improve their effectiveness. For example, if a campaign is not performing well, AI can identify the problem and make recommendations for improvement.**

- Communication through personalization

In a general sense, it is anticipated that the impact of AI on personalization will significantly increase within the marketing context. [77] In another study, Kumar et al. ([2019]) assessed the convergence of AI and personalization, which led to the discovery that personalization is a pivotal factor when it comes to the popularity and adoption of AI. Before the advent of AI, personalization was limited by the quality and volume of the information companies had about their customers and their ability to generate insight and implement the insight.[78] However, with AI, personalization has become internalized in corporations across the world, and it has seen a significant growth in its essence per se. Therefore, AI augmented personalization functions and made it easier for companies to offer improved personalized services. This is a significant difference from the initial years of personalization, when its efforts in marketing were limited based on predefined rules created by experts .[78]

Personalization is an effective way to drive revenue growth, increase customer engagement, and enhance customer satisfaction. According to a survey by Accenture, [91]% of consumers are more likely to shop with brands that recognize, remember, and provide relevant offers and recommendations. In recent years, businesses have recognized the value of personalization in improving customer experience by leveraging artificial intelligence (AI). With advancements in computing, companies can now analyze vast customer data to gain insights into individual preferences, behavior, and needs, enabling them to offer AI personalization in the form of personalized recommendations, tailored products, and customized services meeting each user's unique needs. [79]

Personalization involves tailoring experiences to meet individual customer needs, preferences, and interests.

Personalization is a critical factor in the success of e-commerce businesses. By implementing personalization strategies, businesses can improve customer experiences, increase conversion rates, and gain valuable customer insights. [80]

For every company, shaped by their overall objectives, there are different kinds of AI that could be required for personalization (from the less complex ones to the highly complex ones). The complexity of AI required to deliver personalization in a company will depend on the degree of personalization the company aims to achieve. The much simpler forms of personalization (for instance, mass personalization, as is common in many web applications) would require less complex intelligence. Huang and Rust ([2018]) went further to highlight the

77. Paschen et al., (2019), Artificial intelligence (AI) and its implications for market knowledge in B2B marketing

78. Kumar et al.,(2019), Understanding the Role of Artificial Intelligence in Personalized Engagement Marketing

79. Rohan Singh, (2023), AI Personalization: Challenges and Practical Strategies for Startups

80. By Web Planet Soft, (2023), The Importance of Personalization in E-Commerce: Strategies and Case Study

nature of the complexity of AI required for mass personalization by stating that it can be undertaken by agents possessing analytical intelligence. However, a company that wishes to offer personalized, quality services based on intuition as the form of knowledge would need to apply intuitive intelligence. [81]

Certainly, AI has enhanced a wide variety of personalization applications. They include: automatic generation of advertising copies[82] ; recommender systems[83] prediction of customers' level of satisfaction ,tailored and targeted marketing for generating leads [84], and a host of other marketing functions in early stage companies.

Netcentrix suggests that AI provides the ability to automate calls that entail simple requests or queries, to free up agents' time and also offer intelligent call queuing which allows customers to self-serve, further streamlining communication channels. This optimisation of business communication improves customer experiences, through reducing frustration and allowing quicker access to human agents for customers who really need them and providing quick answers via automation for customers with simple requests.[85]

AI has revolutionized content creation by enabling more efficient, targeted, and dynamic content production. One significant way it has influenced our work is through enhanced data analysis and personalization. AI algorithms can analyze vast amounts of data to understand audience preferences and behaviors, allowing us to tailor content more precisely to our audience's interests. This leads to higher engagement and a more personalized user experience.

Looking ahead,Lata Tewari, Chief Content Officer at Webuters Technologies Pvt Ltd, anticipates AI will play a pivotal role in content curation by automating and refining the process of selecting and presenting personalized content to individual users. AI's ability to learn and adapt will ensure that the content we curate is not only relevant but also timely and contextually appropriate. This will lead to a more intuitive and responsive content ecosystem, where the relevance of content to each user is continuously optimized.

She adds, I see AI contributing to the creative aspect of content creation. With tools like GPT and others, AI can assist in generating ideas, producing drafts, and even suggesting content improvements based on predictive analytics. This will not replace the human touch but rather augment our creative capabilities, allowing content teams to focus on strategy and storytelling while AI handles the heavy lifting of data analysis and basic content generation.

Thus, AI enables startups to analyze large volumes of customer data, including demographics, preferences, and behaviors, to create highly targeted and relevant communication strategies. By understanding individual customer needs and preferences, startups can tailor their messages and content to resonate with specific audience segments, increasing the effectiveness of their communication efforts. Additionally, AI-driven recommendation systems can suggest personalized products or services to customers based on their past interactions and purchase history, leading to higher conversion rates and customer satisfaction. Furthermore, AI-powered chatbots and virtual assistants can provide personalized assistance and support to customers in real-time,

81. Huang & Rust, (2018), Artificial Intelligence in Service,

82. Deng et al, (2019), Smart Generation System of Personalized Advertising Copy and Its Application to Advertising Practice and Research,

83. Zanker et al., (2019) Measuring the impact of online personalisation: Past, present and future

84. Syam & Sharma, (2018), Waiting for a sales renaissance in the fourth industrial revolution: Machine learning and artificial intelligence in sales research and practice

85. Adam Turton , (2023), Embracing the Future Top Opportunities and Challenges

improving the overall customer experience and building stronger relationships. Overall, personalization with AI empowers startups to deliver more engaging, relevant, and effective communication that drives customer engagement, loyalty, and ultimately, business success.

> **Personalization is a critical factor in the success of e-commerce businesses. By implementing personalization strategies, businesses can improve customer experiences, increase conversion rates, and gain valuable customer insights.**

- Communication through Content creation

One obvious area of application for AI in companies particularly startups in early stages is content creation. In some organizations, up to 50% of content is already generated using AI technologies. This includes not only text, but also images and videos. The use of artificial intelligence promises significant improvements in terms of quality and productivity in various business areas. This represents a significant opportunity for marketing specialists, who should now consider how they can integrate AI into their own strategies and processes. 86

Content has emerged as one of the most critical and influential marketing tools [87] and content creation and curation in particular have seen significant influence by adopting AI-powered marketing techniques. [87]

As more content is created and curated every passing hour across almost all media of information consumption, there has been a growing need for content personalization. Indeed, the need for extreme content personalization had emerged from the need to generate automated insights using AI-powered content marketing [88] which has been served by building content recommender systems using narrative science methodologies. [89]

Text generation tools have created a whole new content creation workflow that companies are using to create marketing and sales assets more effectively.[90]

AI also has sped up the content-creation process. From brainstorming to narrowing research, AI has been a great help in quickly developing content outlines and pillars. Although we cannot rely on it fully, it offers a sense of direction, especially when feeling inspired or suffering from writer's block. AI has been helpful in repurposing content and in writing meta descriptions, etc.

Nicole Lee, Content Manager at Remote Company claims that AI will continue to influence the future of content, but not so much that it will replace great expert writers or content strategists. AI will always be a few steps behind in

86. matelso.com, (2024)

87. Kose et al, (2017), Improving Content Marketing Processes with the Approaches by Artificial Intelligence

88. Karimova & Shirkhanbeik, (2019), MARKETING ARTIFICIAL INTELLIGENCE:CREATING THE AI ARCHETYPE FOR EVOKING THE PERSONALITY TRUST

89. Ahmad, 2018; Karimova & Shirkhanbeik, (2019), Artificial intelligence in marketing: A systematic literature review

90. Ethan Crump, (2024), How AI Is Revolutionizing B2B Strategy

identifying what strikes a chord with human readers. However, it will continue to be a big help. As AI progresses, information will become more accurate and speed up the research process for content creation. [91]

> **Text generation tools have created a whole new content creation workflow that companies are using to create marketing and sales assets more effectively**

91. Joshua Long, (2024), How Can AI Help in Content Creation?

2.4 Current Trends in AI Adoption by Startups

The anticipated trends in Artificial Intelligence (AI) and their application within small businesses and startups are indicative of transformative shifts across the global economic landscape.

Xero's research revealed that nearly two-thirds of small businesses (64%) believe emerging technologies such as AI will have an impact on their business in the next five years. [92]

- GENERATIVE AI DOMINATES THE MARKET

Generative AI is poised to dominate the market, with predictions indicating continued expansion and profound implications for businesses. Research by Bloomberg Intelligence (BI) forecasts exponential growth, with the generative AI market projected to reach $1.3 trillion over the next decade, up from $40 billion in 2022. This growth trajectory is attributed to the system's ability to learn from existing data and produce novel data that closely resembles input information. Advanced generative algorithms are anticipated to achieve unprecedented levels of capability, accessibility, and scalability across various domains. Generative AI has demonstrated efficacy in demand forecasting, supply chain optimization, and product development, enabling more efficient processes and improved decision-making. Notable applications include Generative Adversarial Networks (GANs) such as FaceApp and ZAO, which have found utility in entertainment and tailored markets. Moreover, the integration of generative AI in healthcare and customer care has led to significant improvements in patient care, operational efficiency, and client satisfaction through applications like Natural Language Processing (NLP).

In the realm of communication, the influence of AI is profound. Business owners are increasingly relying on AI-driven tools for content creation, marketing, and communications tasks. Solutions such as ChatGPT, Persado, Lex, and Grammarly facilitate tasks ranging from generating content and suggesting topics to researching and writing SEO copy. These tools enhance efficiency and effectiveness in communication endeavors, enabling startups to engage with their audience more effectively and manage their marketing and communication strategies efficiently. [93]

Content tools that have been used recently by startups include:

ChatGPT, Persado, Lex, SEO.ai, CoWrite, Wordtune, Grammarly, NightCafe, Craiyon.[94]

- LLMS POWER UP VIRTUAL ASSISTANTS AND CHATBOTS

Furthermore, Large Language Models (LLMs) are powering up virtual assistants and chatbots, revolutionizing customer interactions. LLM-based digital assistants interact with users on a personal level, mimicking human-to-human interaction and delivering essential information in a personalized manner. These chatbots play crucial roles in categorizing and directing customer inquiries, delivering tailored messages, and offering personalized assistance, thus

92. Alex von Schirmeister , (2024), Four AI trends small businesses should watch in 2024 and beyond

93. Alex Vasilchenko, (2024), Artificial Intelligence Technology, TOP 10 Artificial Intelligence Trends That Will Make a Big Difference in Business in 2024

94. british-business-bank , (2024), AI trends – how AI can help small businesses

enhancing customer satisfaction and engagement. A business doesn't exist without its customers; in a competitive market, anything that can improve customer communication and service is no bad thing.Research in 2020 by Aberdeen found that companies using AI capabilities achieve a 3.5 times greater annual increase in customer satisfaction rates.

One of the most common uses of AI by businesses is customer service chatbots, such as Freshchat, Happy Fox, and Kustomer, to handle routine questions.

This helps relieve the pressure on human customer service agents and more quickly deal with everyday issues.

Additionally, AI-driven chatbots relieve pressure on human customer service agents, enabling quicker resolution of everyday issues and improving overall customer communication and service.

Talking about video content creation to communicate with the customers in a more effective way, A recent TikTok AI trend makes it easy for anyone to turn popular cartoon characters into funny memes. So TikTok users can create memes faster.

Likewise, the recent AI Instagram trend helps creators scale their content. For example, AI-based video editing tools let you cut a long interview into small and engaging short videos to repurpose on Instagram without much manual effort.94

- AI BOOSTS REAL-TIME VIDEO PROCESSING

Moreover, AI is boosting real-time video processing, a trend with significant implications for startups. The integration of AI in data pipeline processing ensures accurate data transmission and minimal video processing latency, enabling efficient handling of real-time video streams. AI-driven enhancements such as background removal and blurring facilitate the creation of high-quality video content, further enhancing communication strategies for startups.

According to market research from the CMI Team, the global Live Streaming Market is expected to record a CAGR of 28% from 2023 to 2032. By 2032, the valuation is expected to reach USD $256.56 billion.

AI boosts internal communication in the Workplace

Startups began to use AI in their workplaces to perform HR, marketing, and production tasks.

HR employees could use AI to source, shortlist, schedule interviews with, recruit, and onboard new candidates with greater efficiency.

Each industry can also find ways to maximize its use of AI differently. Like employing a technology like AI-based computer vision (CV). [95]

“he global Live Streaming Market is expected to record a CAGR of 28% from 2023 to 2032. By 2032, the valuation is expected to reach USD $256.56 billion.”

95. Sydney Go , (2024), 15 Must-Watch AI Trends to Empower Your Business in 2024

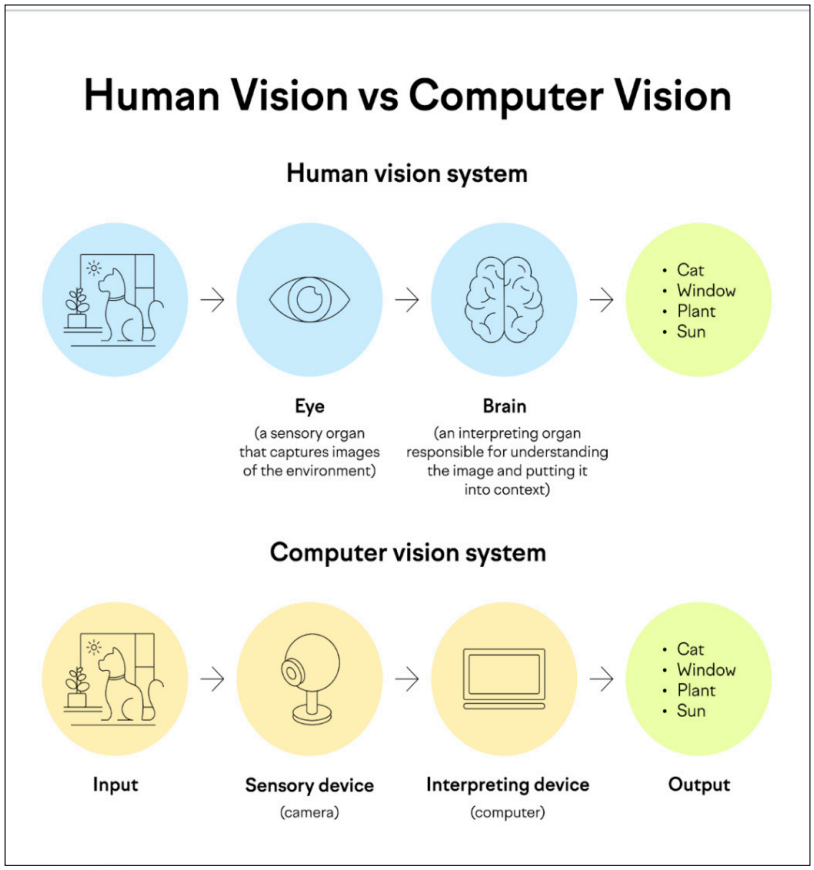


Fig.12
employing a technology like AI-based computer vision (CV) is a current trend that may be done to maximize the use of AI differently.
Bernard Marr
2022

- AI tools help to make HR processes more efficient

Features include writing job advertisements and descriptions, targeting relevant job candidates via social media platforms such as LinkedIn, analyzing CVs to identify the best candidates, chatbots to answer common questions by job candidates and existing staff, automating onboarding processes for new employees and automating training and personal development are done recently by startups using ai tools include:

Fetcher, Paradox, Workable, Freshteam and Bit.ai

- AI Boosts Design

AI is anticipated in UI and UX (user interface and user experience) design and 3D animation to be the top AI design trends in 2024. Incorporating AI into UI and UX design processes will help designers make better data-driven decisions.

For example, designers can use a tool like Lookback to gather detailed data about user actions, heat maps of their mouse movements, and click-through rates (CTRs).

Designers can combine the data with an intelligent AI program like Adobe Sensei to synthesize thousands of user data points into actionable UI/UX design directives.

3D animation designers will also increasingly use AI-based algorithms to create detailed visual effects, environments, and characters.

For instance, 3D animators could use what are known as procedural modeling algorithms to help create content for movies, video games, and more.

These algorithms are essentially a set of rules about content creation that the computer follows.

Rather than spending their time creating new content from scratch, 3D modelers can focus on setting the rules and let the algorithm generate image or video content.

The company DeepMotion uses AI to capture real-life movements and generate virtual content[94]

“AI is anticipated in UI and UX (user interface and user experience) design and 3D animation to be the top AI design trends in 2024. Incorporating AI into UI and UX design processes will help designers make better data-driven decisions.”

2.5 Challenges of AI Integration in startups' communication

The implementation of AI in the early stage businesses' communication has tons of pros, But Despite their functionality, AI tools, and their large-scale implementation, are not without problems. AI technology has many possible uses in business communication and collaboration, but these capabilities need to be considered alongside the known, emerging, and potential challenges. Different researchers highlight the needs to adopt AI technology in the workplace safely by looking beyond performance metrics to risks of data privacy, security, and possible bias.[96] Google massively uses AI technologies, but even their CEO Sundar Pichai called for regulation: "Now there is no question in my mind that artificial intelligence needs to be regulated. It is too important not to. The only question is how to approach it" [97]

Various types of challenges that will be discussed in this chapter include:

Bias
Inaccuracy
Privacy
Proficiency and Use
Professional Standards
Deployment of AI
Effective Data Governance
Collaboration between Humans and AI
lack of emotional intelligence
Replicating Human Creativity and Emotional Understanding

96. Manyika & Sneader, (2018), AI, automation, and the future of work: Ten things to solve for

97. Pichai, (2020), Why Google thinks we need to regulate AI

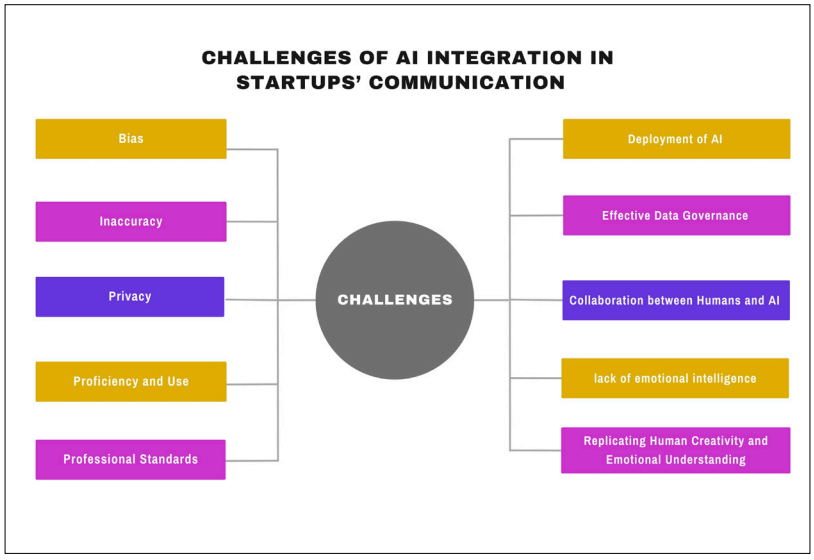


Fig.13 **Several Challenges of AI Integration in startups' communication formulated based on authors' research** Designed by the author 2024

- Bias

One major concern with these technologies is related to bias in decision-making processes. Algorithms mirror human biases and even reinforce them when trained with older data that contains semantic evidence of historic biases. [98] Counter to the idea that AI is objective because it relies on math instead of emotions, AI can reproduce the bias of humans when making decisions. [99] Because AI technologies learn from large amounts of data in an attempt to complete a task, the output of the AI is only as unbiased as the training data on which it learned. If the machine receives biased data, then the machine could produce results more biased than the original data, as it exaggeratedly tries to replicate patterns it found in its training. O'Neil offers the design concerns of transparency, simplicity, and stress-testing (testing the effectiveness of these technologies under a variety of conditions) to counter bias in AI design.

AI bias is particularly damaging because AI technologies lack empathy, mercy, or emotions of any kind. AI cannot feel, use emotional intelligence, or deploy situational awareness. Some people want to use AI for exactly this reason: to offload hard decisions onto the math and justify actions. This desire is misguided, because many situations where people would like to avoid hard choices are particularly where humans are best suited: decisions regarding hiring, firing, and promoting are all complex human choices that require emotional intelligence and intuition, situational awareness, and nuance that current AI cannot approach. AI can sort massive amounts of data with impressive accuracy, but it cannot make decisions with emotional intelligence and intuition. While people in these situations can and perhaps should use AI to provide multiple scenarios and options [100], human judgment can take into account human factors that AI cannot evaluate. Mathematical justification of unemotional decision making is a potentially dangerous pitfall of AI use, especially in business communication situations where emotional IQ is becoming ever more necessary. In this instance, AI may affect business communication negatively, if content decisions are offloaded to AI that can make unethically biased evaluations. Business communication practitioners, scholars, and teachers must keep the possibility of bias in mind as they work with AI.

Thus, AI systems can inherit biases present in the data they are trained on, leading to unfair or discriminatory outcomes. Addressing bias and ensuring fairness in AI algorithms is crucial to prevent reinforcing existing societal inequalities.[101]

98. Caliskan et al., (2017) Semantics derived automatically from language corpora contain human-like biases

99. O'Neil, (2016) Cathy O'Neil. Weapons of Math Destruction: How Big Data Increases Inequality and Threatens Democracy .

100. Gray, (2017), AI Can Be a Troublesome Teammate

101. Kristen M. Getchell1, (2022), Artificial Intelligence in
Business Communication: The Changing Landscape of Research and Teaching

- Inaccuracy

AI's reliance on good data and lack of emotional awareness lead to a trenchant critique of AI: by not comprehending any emotional context or being able to understand heterogeneous data, AI can come to incomplete or inaccurate results .

If the limitations of the tool and the data set are not acknowledged in advance, the results of AI analysis may be no more than very expensively created fallacies. [103] Even with good information, AI is best at predicting; humans are better at judging. Humans and AI working together in combination can work toward eliminating the shortfalls of both AI and solely human decision making. [1]

Yet there are limits even to the success of AI's predictive powers. Facial recognition technology has been a prominent area of error and mistaken prediction in AI development. Despite rapid uptake by many organizations, prominent facial recognition software tools recognize Black and female faces correctly less often than White and male faces . [104] These ongoing inaccuracies have led to some change: In January [2021], HireVue announced that it stopped using facial recognition after significant criticism over possible bias in its proprietary algorithms . Using facial recognition tools comes with significant risk for business communicators due to the known possibility of creating incorrect and potentially damaging communication based on output from facial recognition software. All AI tools must be viewed carefully for their potential effects on communication and collaboration, but facial recognition technology is particularly controversial.

Acting on inaccurate results from AI can produce legal and ethical problems. The legal responsibilities borne by creators of machine learning algorithms are currently unclear, leading to a higher amount of risk in using AI than a known tool. From an ethical standpoint, humans should be just as responsible for AI's outputs as they are for the outcomes of the tools they use in construction settings. Humans, not AI, should be blamed when the AI breaks something, just as humans are blamed when bridge bolts break, not the bolts themselves . [105] However, not everyone sees AI as tools, as we will explain below in the Machines as Teammates section, and this view of AI complicates the legal and ethical norms surrounding responsibility for inaccuracy. Regardless, business communicators using AI tools need to be aware that inaccurate results regarding the outputs of the machine are possible.

Thus, If the AI algorithms aren't relying on accurate data, there can be bias in the outcomes.[106]

- Privacy

Another concern while using AI and ML in startups' communication is data privacy. Data privacy is receiving considerable attention internationally, particularly in the European Union. Because AI technologies require large amounts of data to complete tasks, these large data sets can include information about user demographics, behavior, and communication. In response, there is growing concern with how these data are stored, used, and potentially shared.

102. Shane, (2019), You look like a thing and i love you: how artificial intelligence works and why it's making the world a weirder place

103. Tambe et al., (2019), Artificial Intelligence in Human Resources Management: Challenges and a Path Forward

104. Simonite, 2019), The Best Algorithms Struggle to Recognize Black Faces Equally

105. Vorderbrueggen, 2013), The Best Algorithms Struggle to Recognize Black Faces Equally

106. ecn.co, (2019), BENEFITS AND CHALLENGES OF USING AI IN BUSINESS COMMUNICATION

Recently products such as Amazon's Alexa are facing questions about privacy and security. [107] Rules such as the EU's General Data Protection Regulation, the California Consumer Privacy Act, Data Security Law of the People's Republic of China (adopted June [10], [2021], effective September [1], [2021]), South Africa's Protection of Personal Information Act, South Korea's Personal Information Protection Act, Australia's Privacy Act, and Argentina's Personal Data Protection Act offer varying levels of consumer protection and penalty for companies that use personal data improperly around the world. [108]

Those using AI tools should be aware of the potential dangers of misusing or potentially exposing data in whatever country or region the data comes from.

- Proficiency and Use

A lack of knowledge about how to interact with AI and how to use recommendations provided by AI poses another issue. Employees often are not properly trained in how AI functions and what it actually does. This is sometimes known as the black box problem.[1] At its core, AI provides us with predictions. Yet, many people confuse these predictions with decisions. AI's predictions need to be evaluated by humans and only then serve as behavioral guidance . [1] The most sophisticated models, such as GPT-[3], even pose the challenge that humans can hardly recognize that they are dealing with AI, because GPT-[3] generated texts are difficult to distinguish from human written texts. GPT-[3] is not yet released for public use because of the danger of malicious use of the model e.g., disinformation, fraud in academic essays. [110]

But not only human knowledge about AI is lacking. Many AI tools still lack the necessary sophistication to provide reliable and valid results. This is due to algorithms and data availability. Simple, frequently occurring tasks may be completed successfully, but more complex, specific tasks often lack the necessary training data. One example are chatbots, which still fail to address anything but the most basic inquiries. While they are constantly improving, currently they still lack context-sensitivity, tone-sensitivity, and human intuition. Human communication is nuanced and ambiguous, which is even more difficult to decode for a machine than in human-human interaction. While many researchers want to train AI to respond to human emotions effectively (and ideally more uniformly than humans deal with human emotions), this effort raises complex ethical questions. [109] The ability of AI to accurately and uniformly respond to emotions is still not practical at a large scale but may someday be. Even more distant is AI using emotions, but business communication should be ready with ethical and practical responses (e.g., how to collaborate on a team with a note-taking AI that has emotions?) to this emerging concern.[110]

107. Estes, (2019), The Terrible Truth About Alexa

108. Mazzoli, 2021; Robichaud, (2020), Argentina Personal Data Protection Act (PDPA)

109. Merriam, (2021), If A.I. Only Had a Heart: Why Artificial Intelligence Research Needs to Take Emotions More Seriously

110. Brown et al., (2020), Language Models are Few-Shot Learners

- Professional Standards

Professional organizations have published little regarding the social and ethical concerns of AI implementation. Scholars and authors have developed insights and ethical directives regarding the ethical implementation of AI in businesses, but relevant professional organizations are still working on adopting and encouraging such directives. Currently, professional organizations granting an increased focus on ethics and AI, such as The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems, are focused on the design level. There is comparably little literature regarding the ethics and social responsibility of implementation and use of AI technologies from similar professional organizations. This lack of consistent guidelines around the ethical use of AI technologies in business is concerning, especially as organizations push to integrate these technologies ahead of competitors. While McKinsey & Company includes an "Executive's Guide to AI" on their website that provides a foundational overview of key AI concepts in this AI "race," the authors fail to mention how to ethically implement these technologies (58: Chui et al., 2020). This is a severe oversight, as users must understand the social implications that come with implementing and using these technologies.

To address these and other problems related to the ethical implementation and maintenance of AI technologies, standards are needed. 59: Smith and Green (2018) propose communication standards for leaders and all team members of teams that include AI. These standards build on some of the dilemmas that arise as a result of working with AI, such as lack of trust and emotional intelligence. These proposed communication standards are valuable because they refer to the implementation in teams, not just in design. The continued development of communication standards will only become more valuable as AI technologies enter the workplace in greater numbers. Research and education surrounding the social and ethical implications for AI use is necessary.

- Deployment of AI

One of the major concerns with AI in organizations at present is that many systems are only experimental, and never deployed in production. A pilot AI project is relatively easy to develop and is only meant to demonstrate that the technology is feasible in concept. Deployment, on the other hand, requires a variety of tasks and capabilities that may be in short supply.

Surveys of organizations and market research reports in the US and globally suggest that deployment challenges are widespread with big data and AI. A survey 111 of large financial services and life sciences firms found that firms were actively embracing AI technologies and solutions, with 91.5% of fir-

111. Homas H. Davenport and Randy Bean NewVantage Partners LLC, (2019), Big Data and AI Executive Survey 2019 Executive Summary of Findings Data and Innovation How Big Data and AI are Accelerating Business Transformation

112. DeloitteInsight, (2018), State of AI in the Enterprise, Early adopters combine bullish enthusiasm with strategic investments

ms reporting ongoing investment in AI. But only 14.6% of firms reported that they have deployed AI capabilities into widespread production. In a 2019 global McKinsey survey with the headline “AI adoption proves its worth, but few scale impact,” between 12% (in consumer packaged goods) and 54% (in high tech firms) had at least one machine learning application implemented in a process or product. Only 30% of respondents overall were using AI in products or processes across multiple business units and functions.112

- Effective Data Governance

Implementing an effective data governance framework requires a clear understanding of the startups’ goals, data quality requirements, and existing data management practices. Organizations must establish data quality metrics, set data quality targets, and monitor data quality performance regularly. 113

In addition, startups should invest in data governance technologies such as data catalogs, data lineage tools, and data quality management systems to support the data governance process. By adopting a robust data governance framework, organizations can significantly improve data quality and unleash the full potential of AI in the field of communication.

Furthermore, effective data governance is essential for fostering a data-driven culture within an organization. By promoting collaboration and communication between different departments and stakeholders, data governance helps to break down data silos and facilitates the sharing of data assets. 114

This enables startups to leverage their data more effectively and gain valuable insights into strategic decision-making .115

113. Emmanuel J. Candès, (2009), Exact Matrix Completion via Convex Optimization

114. Vijay Khatri et al, (2010), Designing data governance

115. Zeljko Panian, (2010), Some Practical Experiences in Data Governance

- Collaboration between Humans and AI

AI systems are becoming increasingly more autonomous. As a result, there is a growing need for effective collaboration between humans and AI . 116

This collaboration requires developing AI systems that can understand and adapt to human preferences, communicate effectively, and support human decision making. Technical expertise in human–computer interactions, cognitive science, and explainable AI is crucial for designing AI systems that can seamlessly integrate into human workflows. 117

- lack of emotional intelligence

The findings of the study illuminate the multifaceted challenges posed by AI's lack of emotional intelligence in workplace communications. Participants express concerns regarding the impersonal nature of AI-generated messages, the potential for misinterpretation in sensitive interactions, and the erosion of trust and rapport between employees and AI systems. Moreover, the study identifies organizational factors exacerbating the impact of emotional intelligence deficits in AI, such as workplace culture, leadership practices, and communication norms. The discussion delves into the implications of these findings for organizational dynamics, employee well-being, and the ethical considerations surrounding AI adoption in the workplace. 118

- Replicating Human Creativity and Emotional Understanding

Tehseen Zaiedi claims that, While AI excels at data analysis and pattern recognition, it may struggle to replicate human creativity and emotional understanding. Communications professionals bring a unique human touch to storytelling and relationship-building, elements that AI may not fully grasp. Over-reliance on AI-generated data and insights could also stifle innovation, limiting the potential for unconventional approaches that often arise from human intuition and inspiration. Additionally, AI algorithms may sometimes misinterpret context, leading to communication errors or providing inappropriate responses. Therefore, communications professionals must carefully review AI-generated content to avoid any misunderstandings or misrepresentations.119

116. GUANG-ZHONG YANG et al(2018), The grand challenges of Science Robotics

117. Tim Miller et al, (2019), Explanation in artificial intelligence: Insights from the social sciences
Author links open overlay panel

118. LIANG DENG, (2024), AI is progressing rapidly to help workplace communications in various ways

119. Tehseen Zaidi, (2024), AI in communications: Unveiling opportunities and challenges for communications professionals

Usage of AI in startup's communication (Case Studies)

3

3.1
Case study 1: WalkMe

The leading digital adoption platform is a startup established in 2011 in Tel Aviv, Israel. Walkme is a prominent multinational software-as-a-service (SaaS) company. WalkMe's Digital Adoption Platform (DAP) enables businesses to pinpoint and eliminate digital friction from workflows . Organizations can measure, drive, and strategically optimize their digital presence with WalkMe.

WalkMe experienced 3,000+ hours saved in content creation time, 2.5x improvement in outbound reply rates and 3x more content created by applying an AI tool Called Jasper.

Jasper is an AI tool based company which helps brands Move faster with fewer miscommunications.

Jasper was a game-changer for WalkMe's marketing team. By employing Jasper Brand Voice, the team could strengthen brand authenticity in their content while adhering strictly to brand standards. Corinne further expands on this challenge, "When you have multiple writers writing different things, having the same voice and style across all of your content is really important. With Jasper, we get that continuity plus more volume more quickly."

- Results

Across the sales and marketing teams combined, leveraging Jasper resulted in impressive savings of 3,000+ hours in content creation and a 2X ROI (Return on Investment) relative to cost. Considering the value of the time saved, the WalkMe team can now dedicate more attention to refining strategies and optimizing efforts. Corinne elaborates on this value stating, "Starting is the hardest point. With Jasper, we have the ability to do much more with less because we already have something to work with versus coming up with everything fresh."

Moreover, the incorporation of generative AI in content creation led to a 2.5X improvement in reply rates for outbound prospecting efforts. Jasper's mastery of infusing brand authenticity allowed WalkMe's content to shine.

- The Future of Content Creation

With features like Campaigns, Jasper Chat, and the Extension, Jasper gave the WalkMe team the power to swiftly gather relevant data, rewrite content effortlessly, and craft authentic messages that resonated with their audience. All of this has had far-reaching, transformative effects on the team's content and efficiency.

Today, WalkMe continues to leverage Jasper's capabilities, constantly pushing the boundaries of its content creation process. It's more than just a tool—it's a strategic partner that empowers WalkMe to stay competitive in their market and engaging to their audience.

120. Walkme platform, http://WalkMe

121. Jasper Platform, https://www.jasper.ai/

- Corinne Gabay, Head of Marketing Programs at WalkMe: “Jasper enables us to jumpstart content production at scale, starting with just a sentence. Jasper Everywhere is a game-changer for me. I can brainstorm, build, repurpose, or rephrase content on the fly and write in the document I’m working on, saving toggle time and copy/paste hassle.”



Fig.14
AI’s impact on WalkMe’s communication
Designed by the author
2024

3.2 Case study 2: Grizzle

Grizzle, a digital startup established in 2016, specializes in assisting SaaS and B2B companies in achieving exponential organic growth through content marketing, SEO, and digital PR services. With a dedicated team of skilled writers, strategists, creators, and marketers, Grizzle aims to drive true north metrics for its clients by employing a content marketing methodology focused on specialist research, audience identification, and detailed execution.

With the end goals of increasing organic traffic, widening demographics, and capturing consumers at all stages of the customer journey - the process is one that requires in-depth customer insight and a detailed research process.[122]

- Challenges:

Tom Whatley its Founder & CEO of Grizzle: As you'd imagine, when designing an organic content strategy for high-profile brands, the research process can be arduous. In fact, two of the main issues Grizzle faced boiled down to research and needing a seamless tool or software that could provide the information members of the team required in one place.

Let's go into a bit more detail.

First was the task of locating experts and thought-leaders within particular niches and industries, since It helps give content legitimacy, provides additional insight for content pieces, and provides research and knowledge to include in the articles themselves.

But finding these expertise is always a slow, manual process that involves lots of Googling, and copy-and-pasting.

The second task was identifying publications with which to collaborate and contribute guest blog posts. Grizzle has strict criteria when searching for these publications, so again, this time-heavy process was done manually by members of the team.Then Grizzle decided to go through these tasks, using an AI tool called SparkToro.

SparkToro is a software company that uses AI for market and audience research. Their vision is to Help people do better marketing by making the publications, people, and sources that influence any audience more transparent. All the companies' data comes from three sources: anonymized clickstream, Google SERPs, and public social profiles by using Ai tools to extract this data. [123]

-Results:

The positive results that Grizzle have noticed since introducing SparkToro as an AI tool to their workflow have impacted several parts of their business and the services they offer clients and the communication with them.

Digital PR - finding publications that fit the client's messaging is now easier and quicker. And SparkScore makes it easier for our team to qualify the research.

Content marketing - SparkToro allows the research team to quickly identify key thought-leaders within niche industries, which leads to better quality content and collaborations.

Content amplification - SparkToro allows the team to find social accounts

122.Grizzle platform, https://www.grizzle.io/

123. sparktoro platform, https://sparktoro.com/

and publications that are ideal for networking with and sharing client content with. Again, SparkToro has helped with qualifying these contacts.

Link Building - SparkToro's partnership with Hunter.io means Grizzle can find media contacts, and their contact information, all in one place. SparkToro eliminates the need to use multiple tabs and multiple spreadsheets.

Audience building - SparkToro helps Grizzle identify hashtags, topics, and ideas easier and quicker than before.

In short, SparkToro has helped Grizzle speed up their processes, conduct more efficient research, and identify key contacts easier and effortlessly.

Our clients often say, "Wow, that was quick" when gaining traction for outreach and digital PR efforts.

- Communication Improvement:

The integration of SparkToro into Grizzle's workflow resulted in improved communication with clients and enhanced service offerings. Clients have expressed satisfaction with the speed and effectiveness of Grizzle's outreach and digital PR efforts since integrating SparkToro, demonstrating the significant impact of AI tools on streamlining operations and improving communication within the company and with clients.

OddBytes, a cutting-edge, multi-venture technology startup founded in 2022, focuses on enhancing the connection between consumers and brands through personalized experiences. By offering tailored experiences, brands can effectively promote their products and services to their target audience, while online users can discover new stores, products, and receive relevant recommendations.

This startup used an AI tool called Beautiful.ai to Improve its Employee Communication and Engagement.

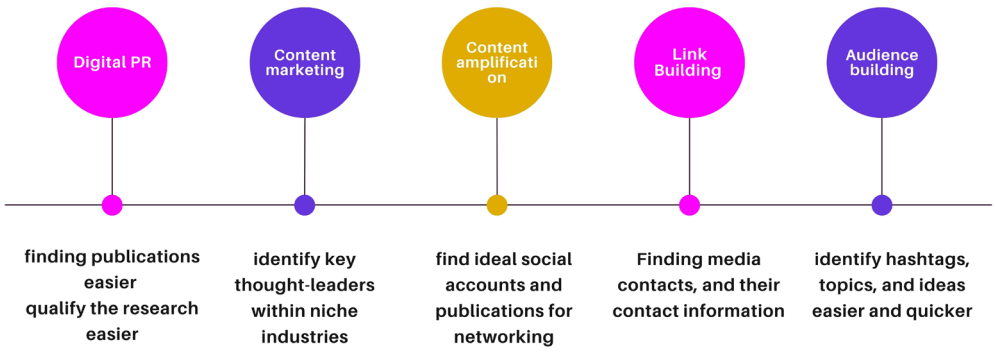


Fig.15
AI's impact on Grizzle's communication
Designed by the author
2024

3.3 Case study 3: OddBytes

OddBytes, a cutting-edge, multi-venture technology startup founded in [2022], focuses on enhancing the connection between consumers and brands through personalized experiences. By offering tailored experiences, brands can effectively promote their products and services to their target audience, while online users can discover new stores, products, and receive relevant recommendations. [124]

This startup used an AI tool called Beautiful.ai to Improve its Employee Communication and Engagement.

Challenges:

With 135 employees dispersed globally, OddBytes faced challenges in facilitating effective communication and collaboration across the organization. Traditional methods of communication, such as lengthy email chains, hindered productivity and efficiency. Additionally, language barriers posed obstacles for individuals like Dani, whose second language is English, in organizing and presenting their thoughts effectively.The CEO of OddBytes says, “Presenting things in Slack is so much more helpful than having to sort through my 600 notification emails.” Beautiful.ai’s Slack integration removes the clutter of lengthy email chains and unlocks a more efficient way to collaborate on decks throughout the business.

Result:

To address these challenges, OddBytes implemented Beautiful.ai, an AI-powered presentation tool, to improve communication and engagement within the organization. The integration of Beautiful.ai with Slack streamlined collaboration on presentations, allowing team members to assign slides, provide real-time feedback, and reduce reliance on email communication. For individuals like Dani, Beautiful.ai’s DesignerBot rewrite feature proved invaluable in organizing thoughts and refining copy, thereby enhancing individual communication effectiveness. Moreover, Beautiful.ai’s visually appealing templates and emphasis on concise, impactful presentations transformed the way information was conveyed, leading to increased employee engagement and trust in the narrative presented. Town hall meetings at OddBytes witnessed a shift towards more interactive and engaging presentations, fostering greater attention and participation from colleagues and peers.

Communication Improvement:

The adoption of Beautiful.ai significantly improved communication within OddBytes, fostering better collaboration, engagement, and trust among employees. Integrating Beautiful.ai with Slack facilitated seamless collaboration on presentations, reducing cluttered email communication and enhancing overall efficiency. Dani’s experience exemplifies how Beautiful.ai’s features, such as the DesignerBot rewrite feature, empower individuals to communicate ef-

124. beautiful.ai platform, https://www.beautiful.ai/

fectively, particularly in multilingual environments. Moreover, Beautiful.ai's emphasis on visual storytelling and interactive presentations has transformed the way information is presented, leading to increased engagement, attention, and participation during meetings and presentations. Overall, Beautiful.ai has revolutionized OddBytes' presentation style, fostering a more interactive, engaging, and professional communication culture within the organization.

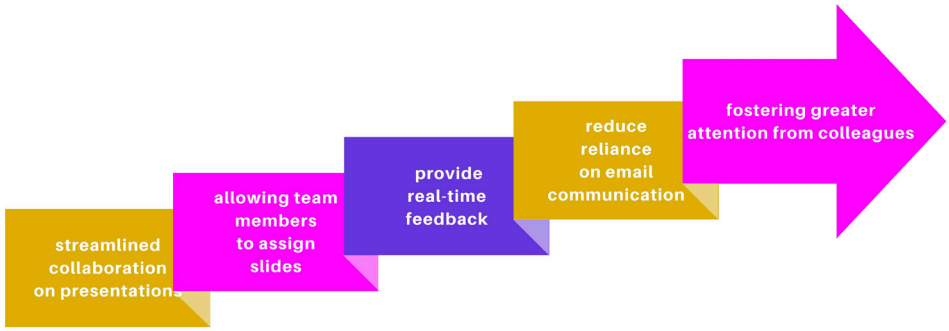


Fig.16
AI's impact on OddBytes's communication
Designed by the author
2024

3.4 Case study 4: Elevatus

Elevatus, headquartered in Riyadh, Saudi Arabia, operates as a talent management platform specializing in the pre-HR cycle of recruitment, hiring, and onboarding. Within their Innovation department, they conceive and develop intelligent systems aimed at streamlining clients' hiring processes by mitigating complexity. [125]

Challenges:

Encountering a challenge during the developmental phase, the team embarked on creating products preceding the era of MLOps, with a limited number of models operational in production. Yanal Kashou, Chief Innovation Officer at Elevatus, recalls the early stages of model training lacking an observability layer, resulting in cumbersome processes to provide evidence when requested by colleagues. Recognizing the necessity of a mature observability layer prior to large-scale training operations and stakeholder engagement in AI system auditing, Elevatus sought a model tracking solution meeting specific criteria outlined by Yanal.[126]

The startup decided to go through these challenges by leveraging ai tools through using Neptune.

Results

A notable improvement observed with Neptune's adoption was in the realm of visualization, an area previously challenging for the Elevatus team. Prior evaluations of tools such as MLflow and TensorBoard highlighted their inadequacy in providing the desired observability dashboards and flexibility required by the team. In contrast, Neptune's customizable interface allowed for tailored views and dashboards, enabling precise monitoring of pertinent metrics.

As a result of Neptune's implementation, Elevatus achieved notable advancements in model performance, evidenced by an average Mean Squared Error (MSE) of ~0.043 and an average Mean Absolute Error (MAE) of ~0.16 across a substantial number of trials. This enhancement in performance, particularly in predicting personality traits from multimedia data, can be attributed to Neptune's facilitation of comprehensive parameter analysis and comparison, empowering Elevatus to optimize their models effectively.

In conclusion, the integration of Neptune into Elevatus's workflow has not only addressed existing challenges but also significantly improved communication within the company by providing a centralized platform for tracking, visualization, and collaboration.

Communication enhancement:

Communication enhancement occurred through Elevatus's adoption of AI tools, specifically Neptune, to streamline their workflow and improve colla-

125. elevatus platfor, https://www.elevatus.io/

126. neptune.ai Platform, https://neptune.ai/

boration within the company. By integrating Neptune into their model tracking and visualization processes, Elevatus gained a centralized platform for tracking model performance, comparing experiments, and sharing insights with team members. This facilitated transparent communication and knowledge sharing, as team members could easily access and interpret data, leading to more informed decision-making. Additionally, Neptune's customizable dashboards allowed Elevatus to tailor their monitoring solutions to their unique requirements, enabling focused discussions and problem-solving sessions. With improved visibility into model performance and streamlined collaboration, Elevatus was better equipped to communicate the effectiveness of their AI-driven solutions both internally and externally, ultimately enhancing their ability to meet client needs and drive business success.

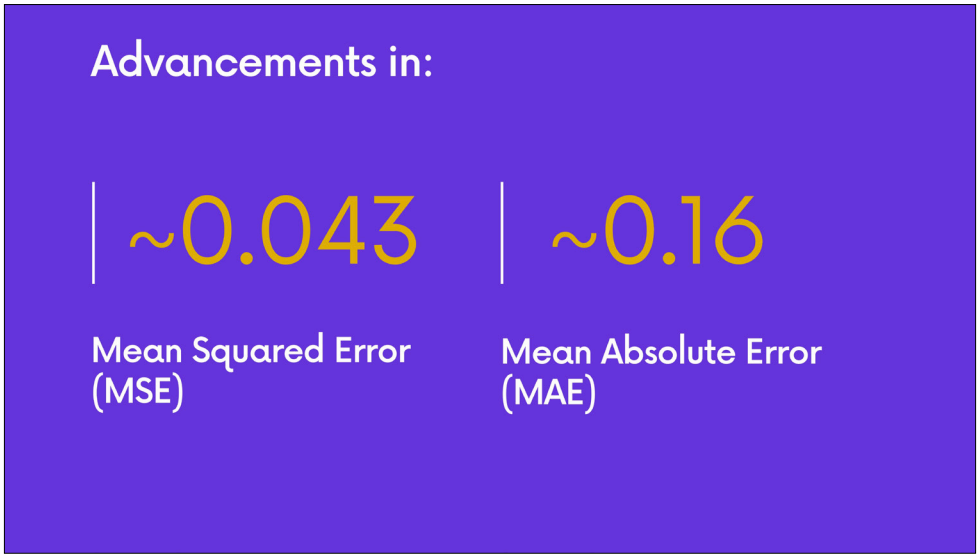


Fig.17
AI's impact on Elevatus's communication
Designed by the author
2024

3.5 Case study 5: Loopio

Loopio, an RFP automation software provider, empowers users to respond to Requests for Proposals (RFPs) swiftly through a blend of response libraries, smart search features, duplication monitoring, and automated review cycles, based in canada.

This startup decided to use Drift as an AI tool to go through their challenges.[127]

Challenges:

Despite their prowess in automation, Loopio recognized a significant opportunity to refine their digital engagement strategy. While employing Drift on their website facilitated visitor engagement and meeting bookings through live sales agents and automated chatbots, they faced challenges in effectively targeting users based on their intent. Existing playbooks, including a generic "Engage All" approach and page-based targeting, proved suboptimal, resulting in missed prospects and increased administrative burden.

Solution:

To address this challenge, Loopio partnered with Lift AI, a tool capable of predicting buyer intent scores for website visitors in real-time. By leveraging Lift AI's pre-trained artificial intelligence model, Loopio could identify visitors demonstrating strong buyer intent and tailor engagements accordingly. This integration enabled Loopio to refine their Drift implementation, leading to enhanced visitor engagement and improved sales team efficiency.

Enhanced Engagement Using Intent Data:

Loopio implemented playbooks exclusively targeting high-intent traffic, connecting such visitors directly with live sales agents during business hours, irrespective of their current website page. Additionally, they deployed playbooks to engage medium- and low-intent visitors, either qualifying them further for sales engagement or providing nurturing resources based on their intent scores. This approach ensured that sales agents focused their efforts on converting high-intent visitors while efficiently managing medium- and low-intent traffic.

Results:

The transition from a page-based to an intent-based approach yielded remarkable outcomes for Loopio. Within one month of implementing Lift AI and high-intent playbooks, the company experienced a substantial increase in meetings booked, opportunities identified, and revenue generated. Specifically, they observed a ^{733}% increase in meetings booked, an ^{800}% surge in opportunities, and a ^{251}% rise in revenue.[128]

Communication Improvement:

This strategic shift not only led to tangible business outcomes but also facilitated improved communication within the company. By effectively targeting visitors based on intent, Loopio enhanced engagement with prospects, optimi-

127. loopio platform, https://loopio.com/

128. drift Platform, https://www.drift.com/case-studies/loopio/

zed sales team efforts, and fostered alignment between marketing and sales functions. Additionally, the implementation of intent-based playbooks enabled Loopio to tailor communications to visitors' specific needs, resulting in more meaningful interactions and increased conversion rates.



Fig.18
AI's impact on Loopio's communication
Designed by the author
2024

Design process for a startup platform

4

4.1
Crafting Ideas

The initial phase of this study involved utilizing Google Trends and conducting a comprehensive market observation, which revealed a significant demand among Italian consumers for at-home beauty services. Italy, known for its aging population, has a substantial segment of elderly individuals who prefer the convenience of receiving beauty treatments at home rather than traveling to a salon. This demographic trend underscores the potential for at-home beauty services tailored specifically to the needs of older adults.

Additionally, Italy's prominent position in the global fashion industry, coupled with its annual influx of models and influencers, further amplifies the demand for accessible beauty services. These professionals often require immediate and convenient beauty treatments, particularly when they are engaged in work-related travel. Italy's status as a major tourist destination also creates a need for on-demand beauty services for travelers, especially those staying in hotels who prefer to receive beauty treatments in their rooms for reasons of convenience and efficiency.

Market research indicated a notable gap: the absence of a mobile application offering on-demand beauty services at home in Italy. This gap presents a unique opportunity to cater to the aforementioned target audiences—elderly individuals, fashion industry professionals, and tourists. Consequently, this thesis focuses on developing a startup idea that addresses this unmet need by creating a mobile application dedicated to providing at-home beauty services across Italy.

4.2
Market research

In the initial phase of improving startup communication, it was imperative to identify the specific beauty services needed within the country, the target audience, and how the demand for beauty services varies between men and women. This foundational information was crucial for building the brand effectively. To gather these insights, the AI tool SurveyMonkey was utilized. SurveyMonkey, leveraging AI, generated a comprehensive survey based on a predefined prompt. The survey targeted a diverse demographic, encompassing various ages, genders, and social strata. The survey started using a prompt as can be seen in figure 19, and the survey is what can be seen in continue.

> **Prompt: Our product is an On demand Beauty service at home mobile application for Italy. We want to learn more about which beauty services are most popular in Beauty industry of Italy, how much italian men and women are willing to pay for the beauty services monthly, Include demographics like age, gender, income level, and anything else that influences purchasing decisions.**

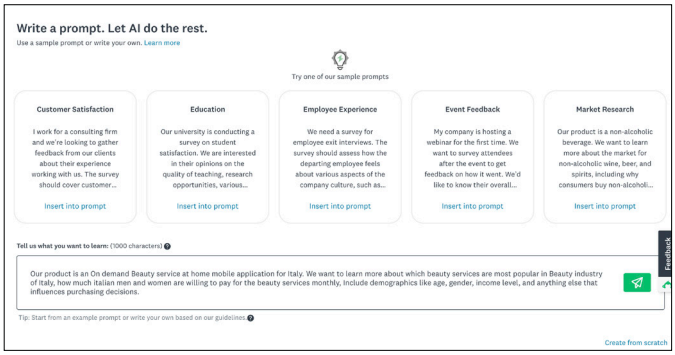
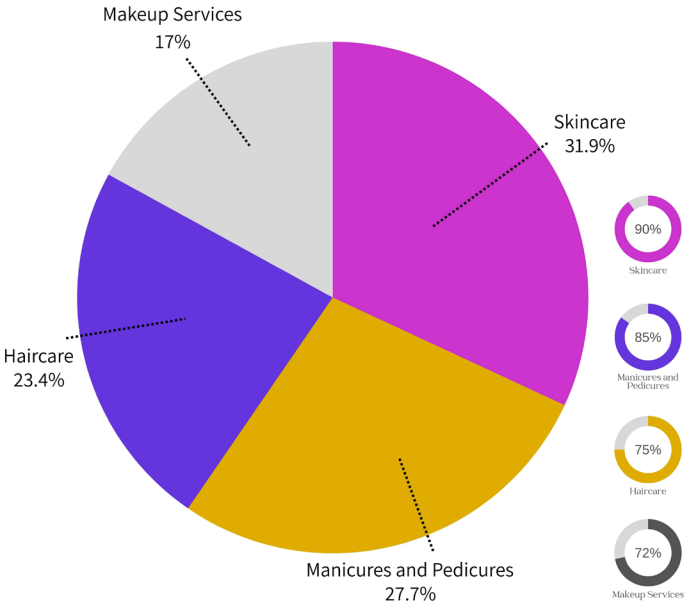




Fig.19
screenshot an AI platform and prompting ot to make a survay for market research and the survey created for the market research
SurveyMonkey
2024

The collected data provided valuable insights, as illustrated in Figure 20 and 21.



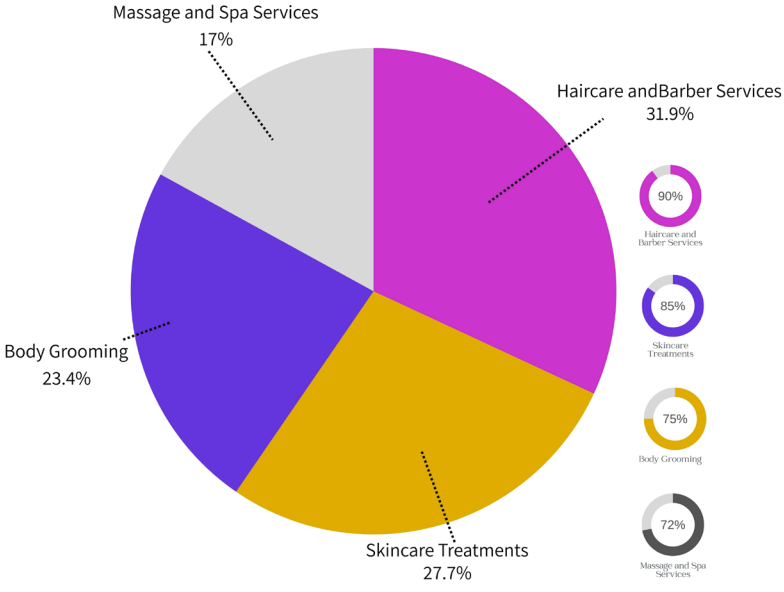


Fig.20 nad 21
Collected data on the different needs o beauty services in Italy for male and female
Designed by the author
2024

4.3 Visual communication design process

This essay primarily examines the relationship between the use of AI and communication within startups. However, the final project places greater emphasis on the visual communication process. As illustrated in Figure 22, various stages in the development of a startup—from naming and logo creation to color palette design, illustration, and mobile UI application design—were explored in this process. Figure 22 details the interrelationships between these stages and their impact on communication.

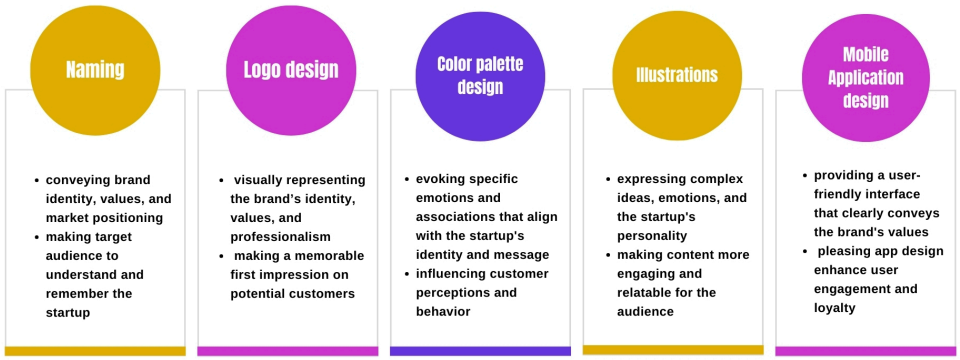


Fig.22
The relationships between different stages of creating a startup visual communication and their impact on communication
Designed by the author
2024

- Naming:

For naming process for this startup I asked GPT 3 its ideas based on creating a name for this brand and it came to some names, which I finally chose the name “CasaBellezza”, but asking locals so I continued asking Gemeni, and finally the name “GlamOnGo” was chosen, as can be seen in figure 23.

“Prompt: Suggest some Noble names for an “on demand beauty services at home” mobile application in italy.”

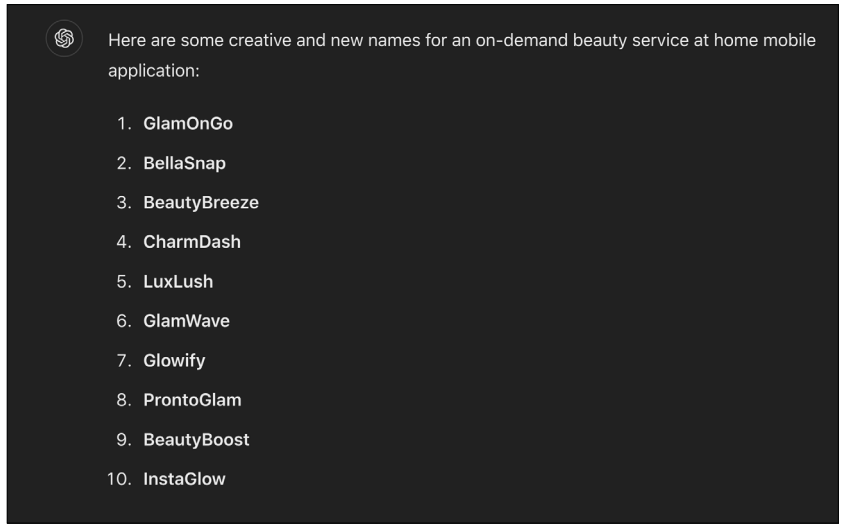


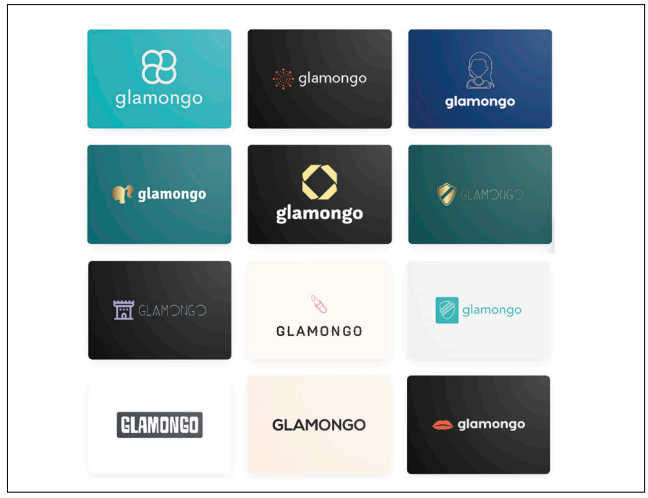
Fig.23
screenshot of an AI platform and prompting it to suggest names
Chatgpt 3
2024

- Logo design:

Several AI-based websites offer tools for creating logos entirely using artificial intelligence algorithms. However, the question arises: are the outcomes of these tools professional enough for initial use in establishing a business or a brand? To investigate this, a series of experiments were conducted.

- LOOKA

Initially, the AI platform of LOOKA was utilized, as the results can be seen in figures 24 and25.



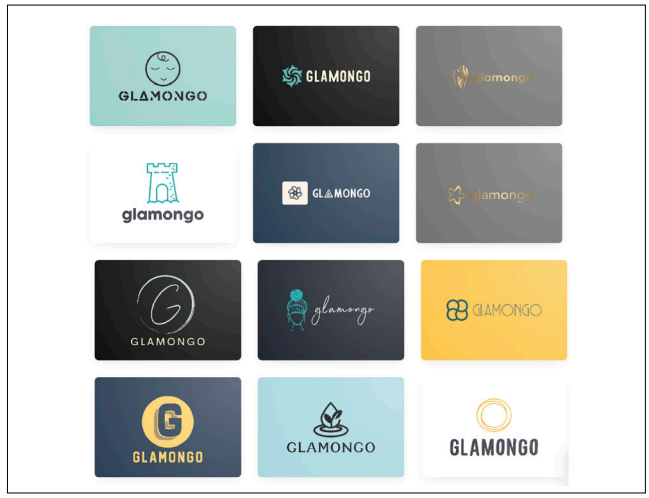
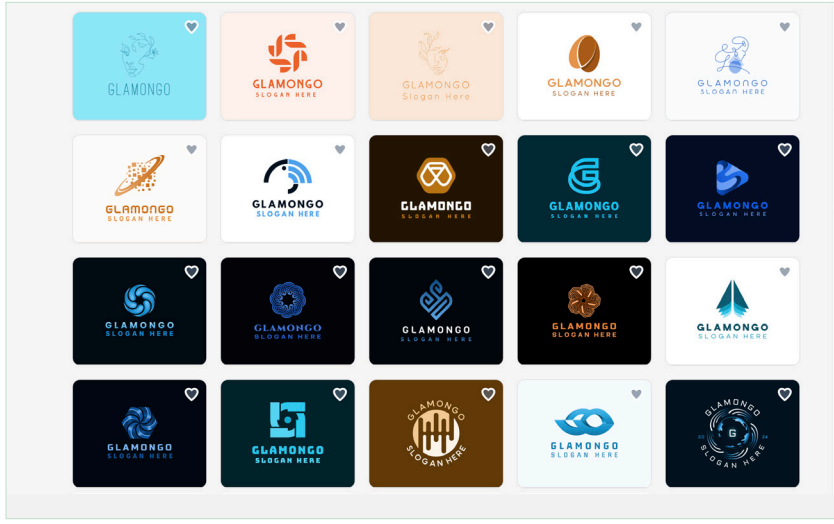
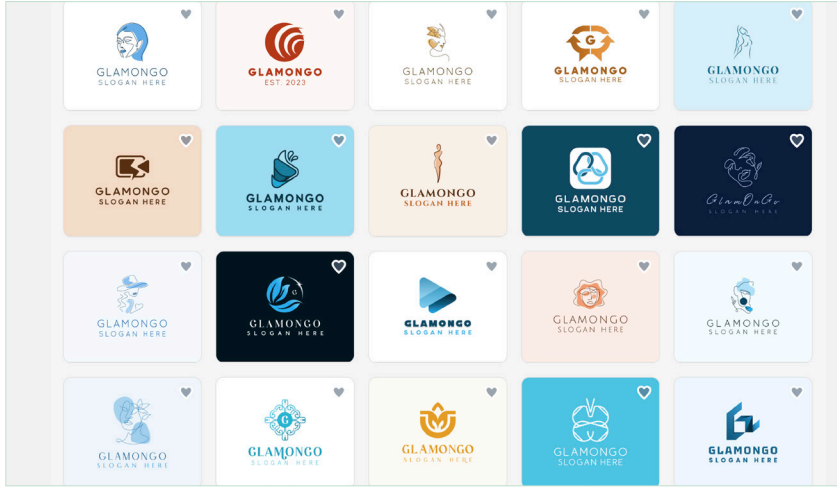


Fig.24 and 25
Logos created by an AI platform
Looka
2024

Subsequently, another attempt was made using ai tool of Design.com, the results can be seen in figures 26 and 27.





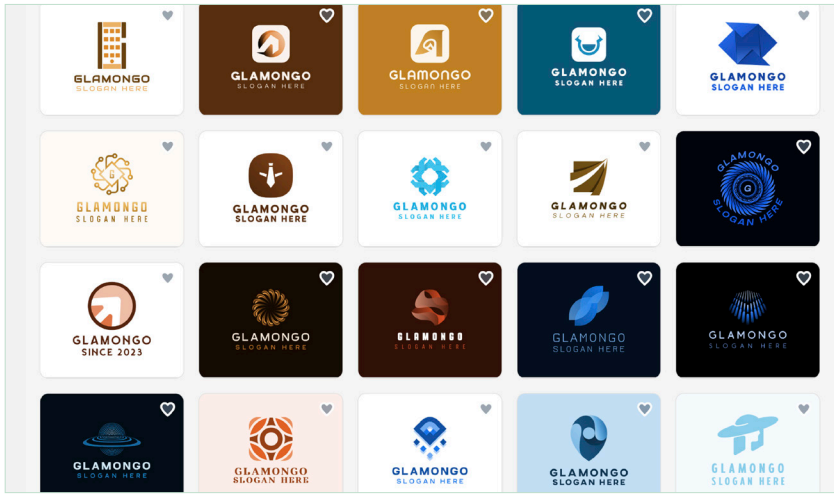


Fig.26 and 27
Logos created by an AI platform
Design.com
2024

Finally, Logomaster AI tool was explored, as can be seen in figure 28.

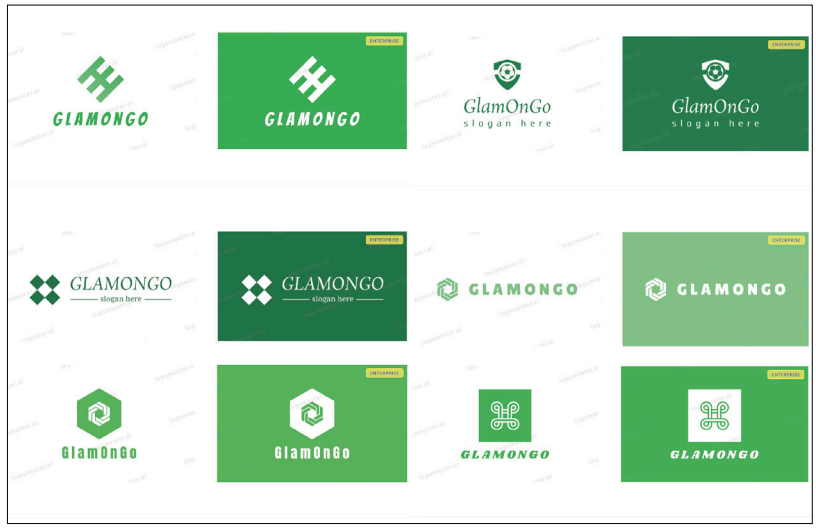


Fig.20
screenshot oan AI platform and prompting itLogos created by an AI platform to suggest how to make logo
Logomaster.ai
2024

Fig.28
screenshot oan AI platform and prompting it to suggest how to make logo
Logomaster.ai
2024

Consequently, as a last resort, input was sought from GPT-3 to identify the fundamental characteristics of the brand's logo, and attempts were made to design the logo based on its suggestions.communicative plan for communication with investors as can bee seen in figure 29.

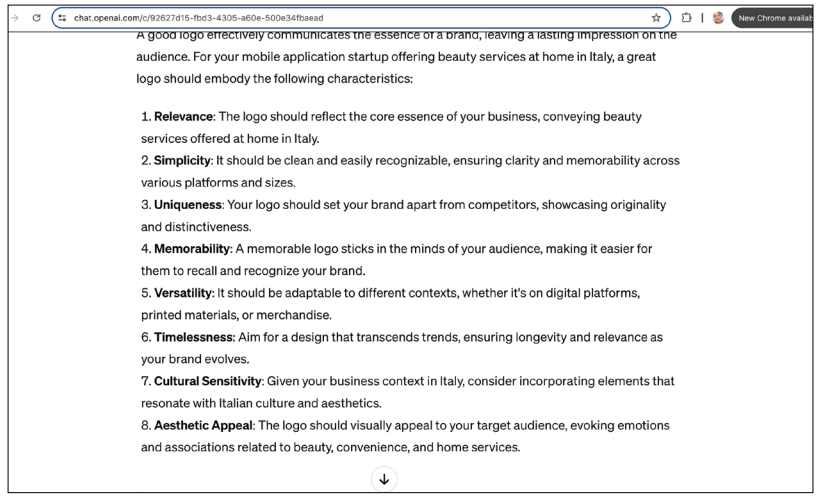


Fig.29
screenshot oan AI platform and prompting it to suggest how to make logo
Chatgpt 3
2024

Based on those suggestions the final logo of the start up had been created using manual and traditional techinques in Adobe Illustrator, as can be seen in figures 30 and 31.

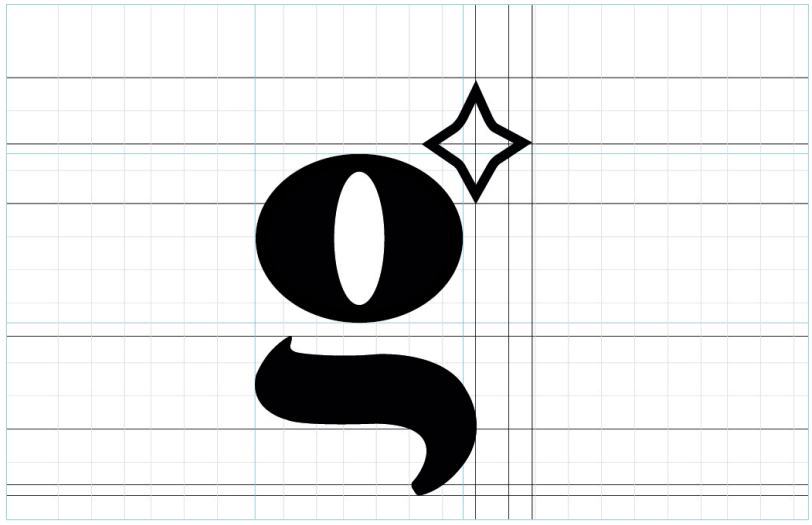






Fig.30 and 31
GlamOnGo Logo
designed by Author
2024



Fig.31 **GlamOnGo Logo** *designed by Author 2024*

- Color palette design:

In the pursuit of designing a color palette for the project, two AI-based websites were employed.

Initially, the AI tools provided by Canva were utilized, utilizing two images deemed reflective of the brand identity. However, the outcome yielded was not deemed satisfactory, as the results can be seen in figure 32

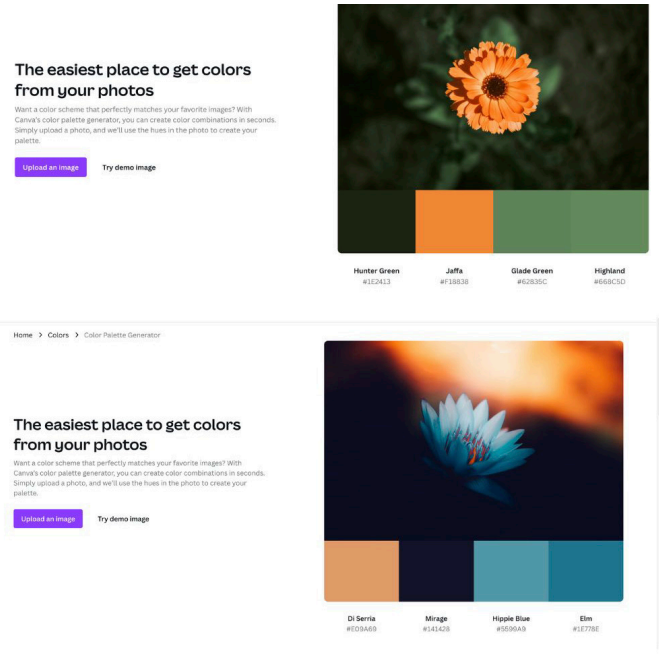


Fig.32 **Suggested color pallette using AI tools** *Canva 2024*

Subsequently, seeking to discern the most suitable colors representative of the brand, Gemini software was consulted. Despite the recommendation of light blue, which was attempted but found unsuitable, the decision was made to explore turquoise blue, To refine the color palette based on this choice, the AI tool offered by the website COOLORS was utilized.

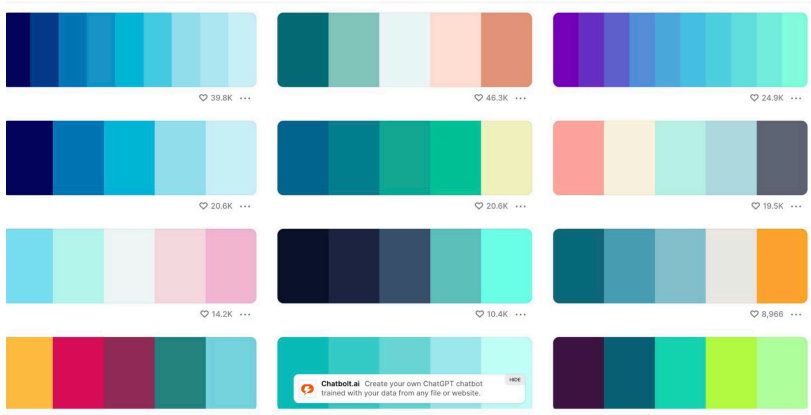


Fig.33
Suggested color pallette using AI tools
Coolors
2024

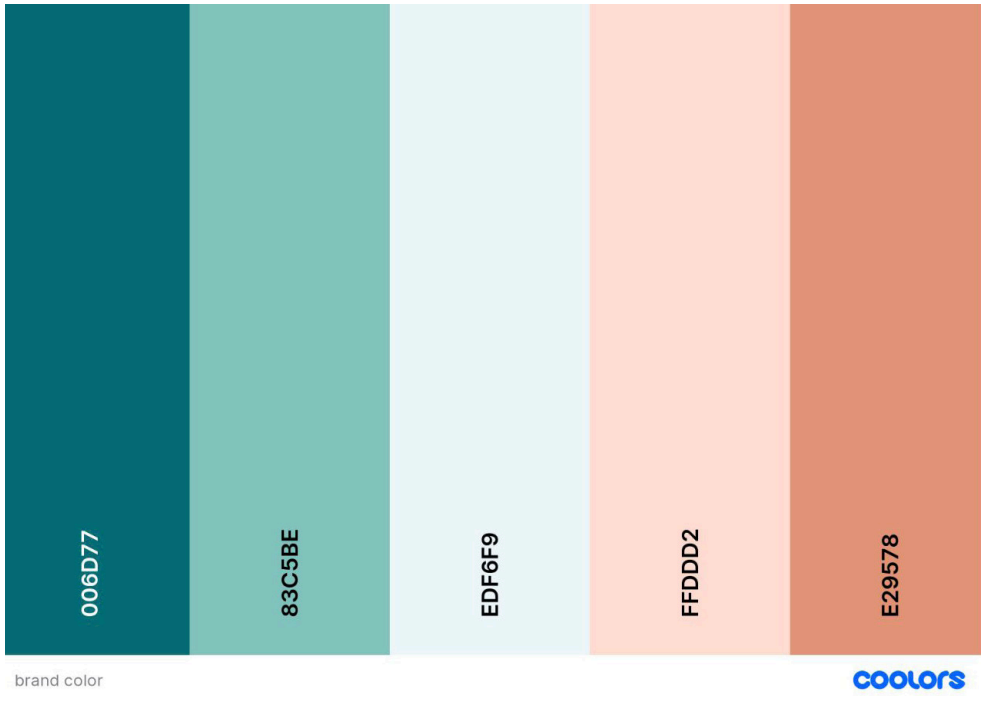


Fig.34
final result of generated color pallette using AI tools
Coolors
2024

- Illustrations:

In the pursuit of designing Illustrations for the project, co pilot ai, an AI-based platfrom was employed as the results can be seen in figure.

> **Prompt: create a beauty service related illustration based on this image, please make the illustration shown at home**

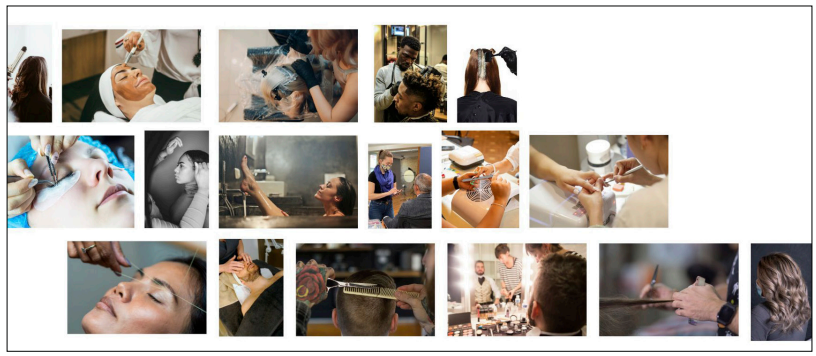


Fig.35 **Refrence Photos to create illustrations** *Unsplash 2024*

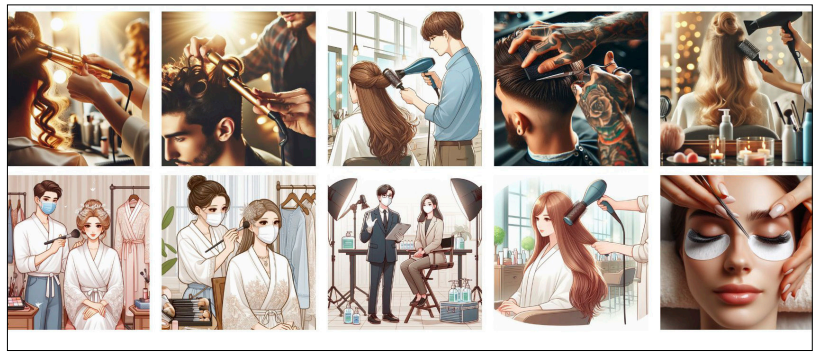


Fig.36 **illustrations generated by AI** *Gemini 2024*

- Mobile Application UI design :

In the endeavor to develop the mobile application's design, Gemini software was consulted to delineate an optimal UI mobile application design suited for the platform in question.

First, gemini ai had been asked for the mobile application design charecters considering the target audience and other factors of the startup. the result was mostly suggestions about Home page , Booking flow, Service provider profiles and Overall design as can be seen in figures 37, 38, 39 and 40.

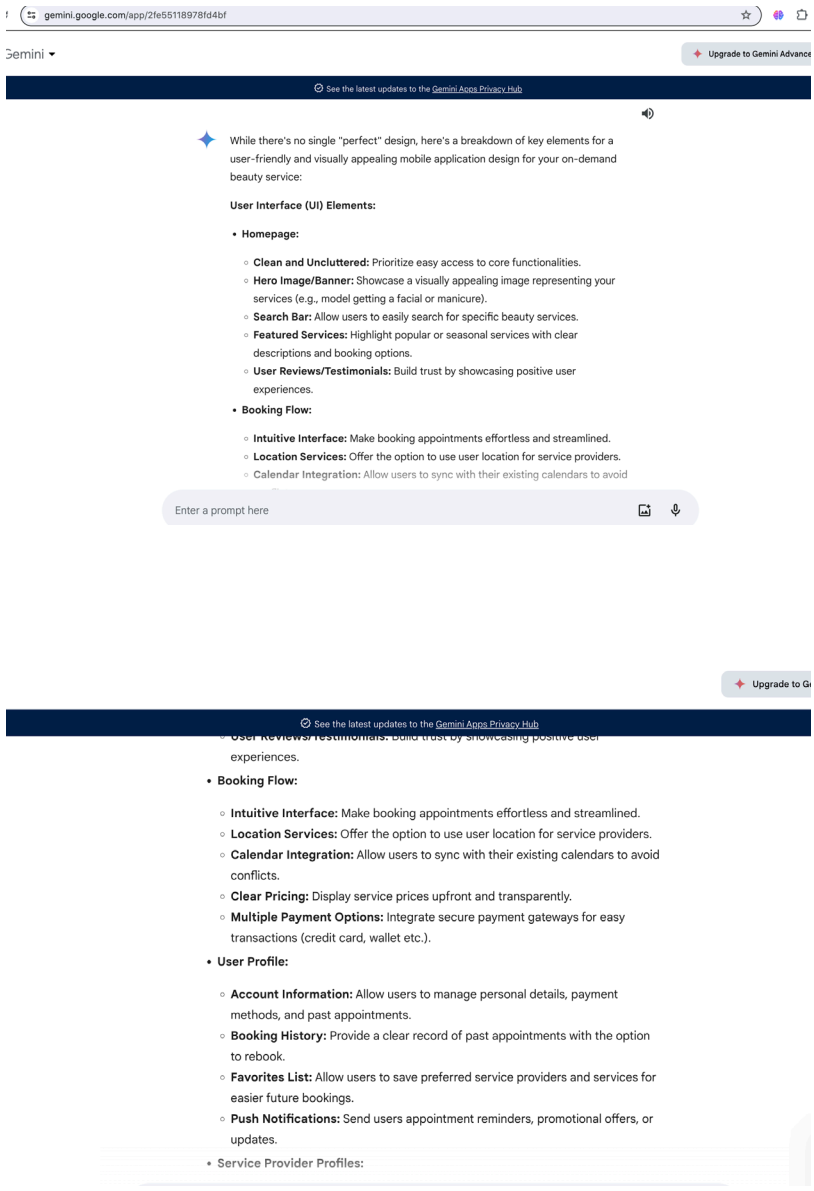
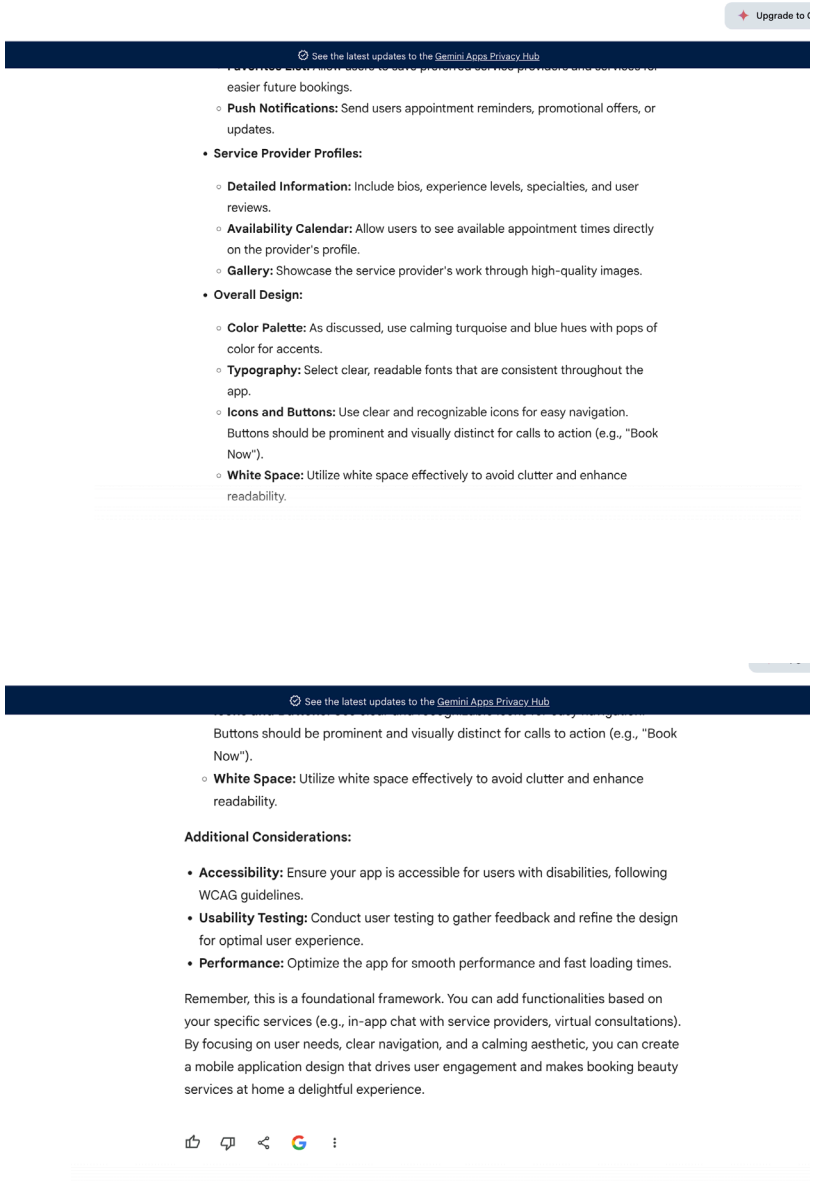
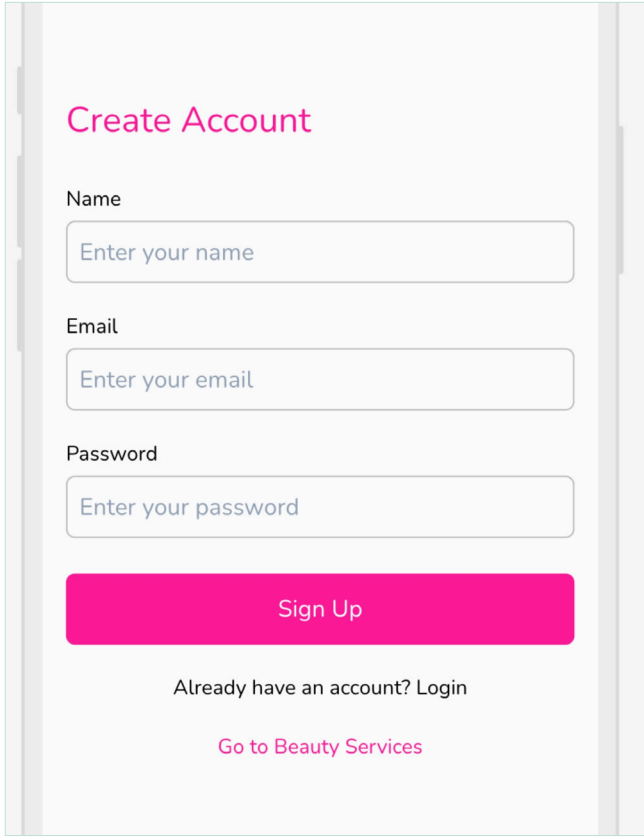


Fig.37 to 40
Suggestions of AI to create UI of startups's mobile application
Gemini
2024



And then an AI based platform called Uizard.io had been used to create some UI designs, although the result could not be used as the final mobile application, the pages were inspirational for the further designs that had been done by me in Figma. the Ui pages created by AI tools can be seen in figures 41 and 42. By considering all these suggestions the final product as the mobile application had been designed, as can be seen in figures 43 to 48.



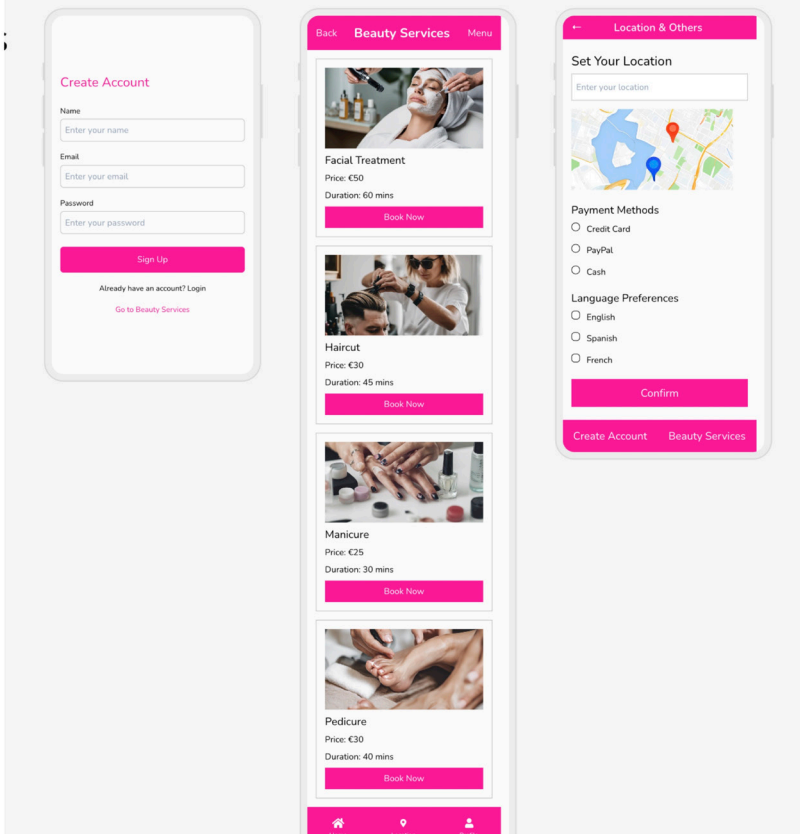


Fig.41 and 42
Ui pages of the startup's mobile application designed using AI tools
Uizard.io
2024

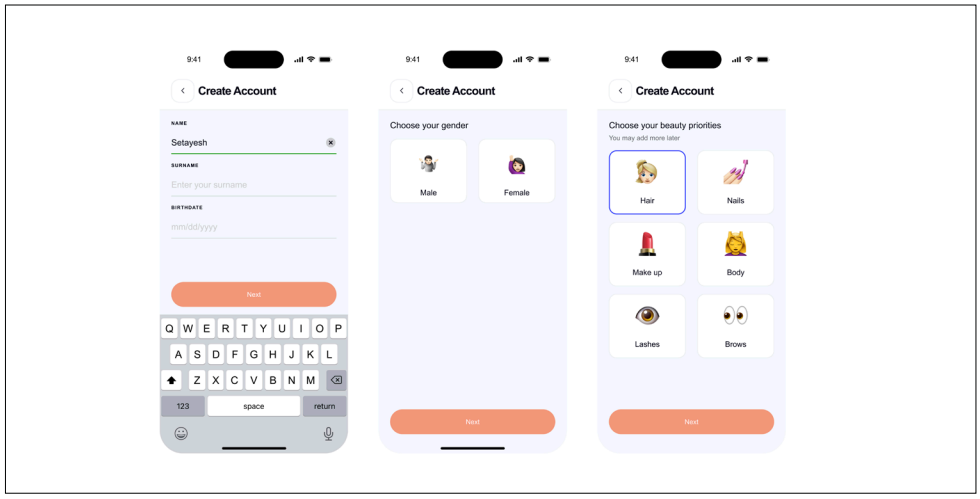


Fig.43
Create Account UI page
ui design of the mobile application
created by the auther in Figma
2024

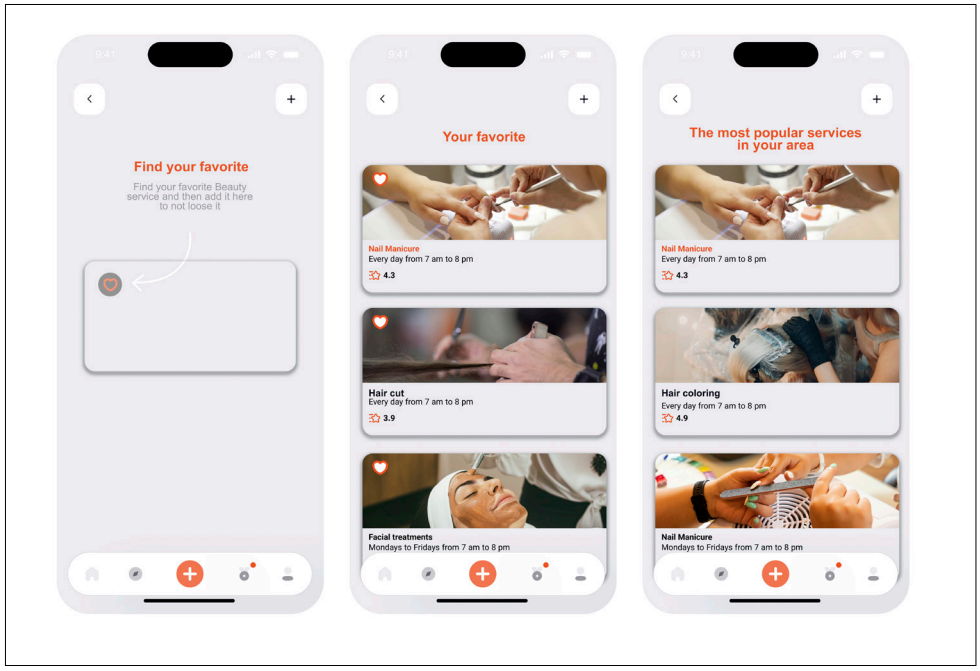


Fig.44
favorites UI page
ui design of the mobile application
created by the auther in Figma
2024

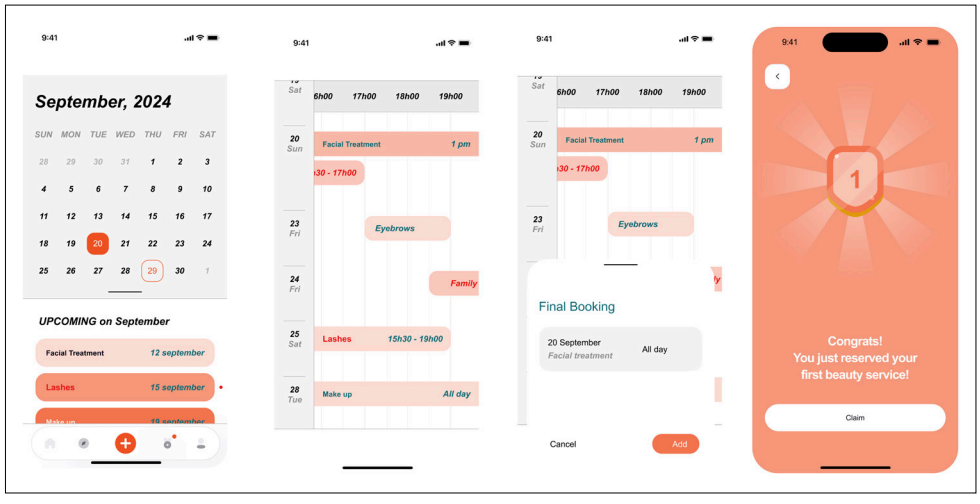


Fig.45
location UI page
ui design of the mobile application
created by the auther in Figma
2024

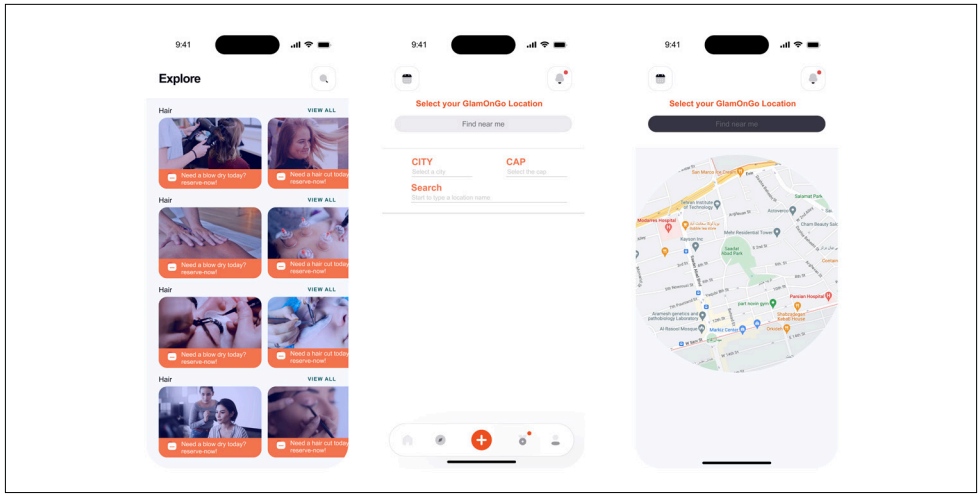


Fig.46
Map UI page
ui design of the mobile application
created by the auther in Figma
2024

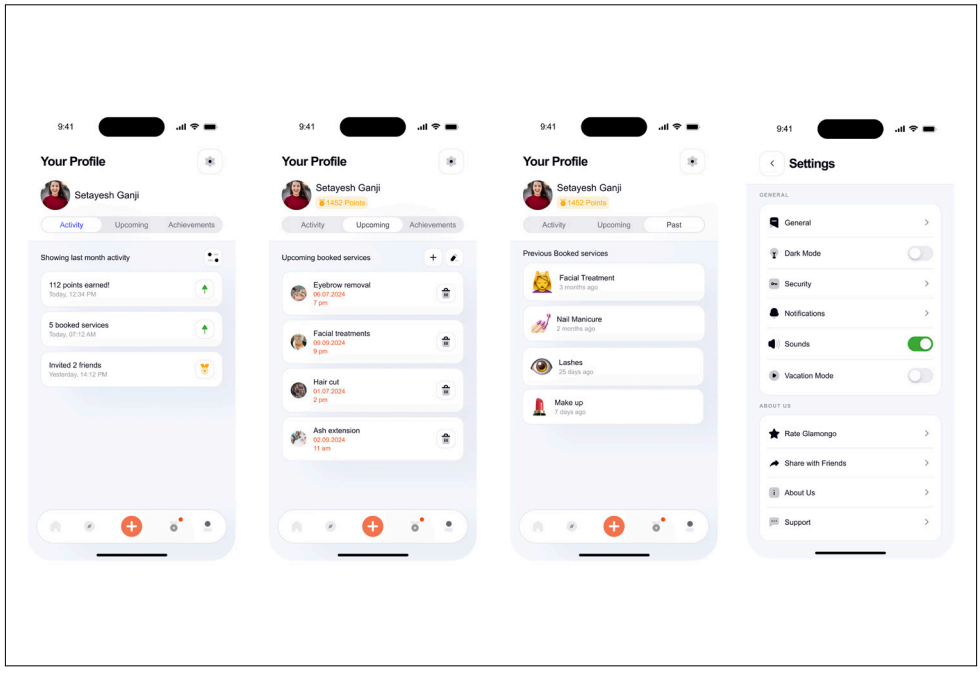


Fig.47
Profile & Settings UI page
ui design of the mobile application
created by the auther in Figma
2024

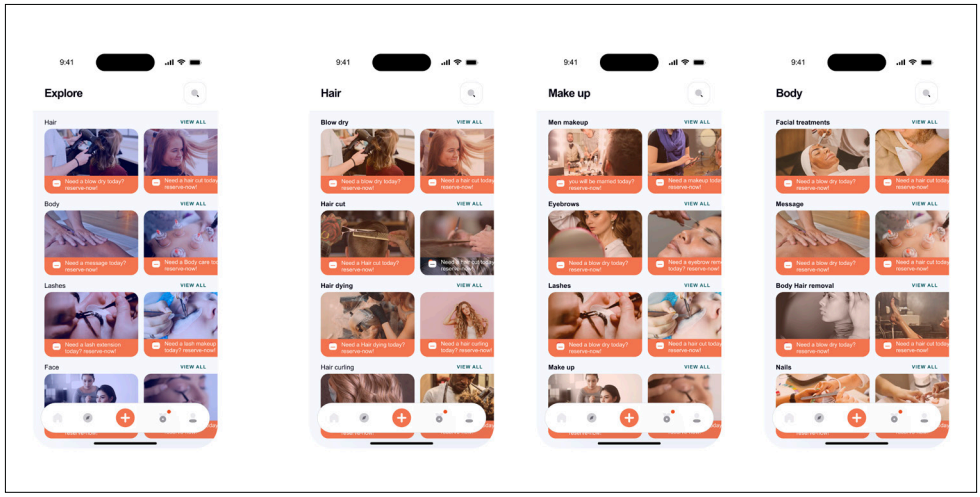


Fig.48
reserved UI page
ui design of the mobile application
created by the auther in Figma
2024

Finalyy it is critical to mention, Figure 49 lists all the AI platforms utilized in the design process and specifies the methods each platform employs to generate results using AI from textual prompts to visual prompts, while some AI tools use Keywords only to create results, meanwhile some of them use only other samples, and some of them use the combination of these methodes.

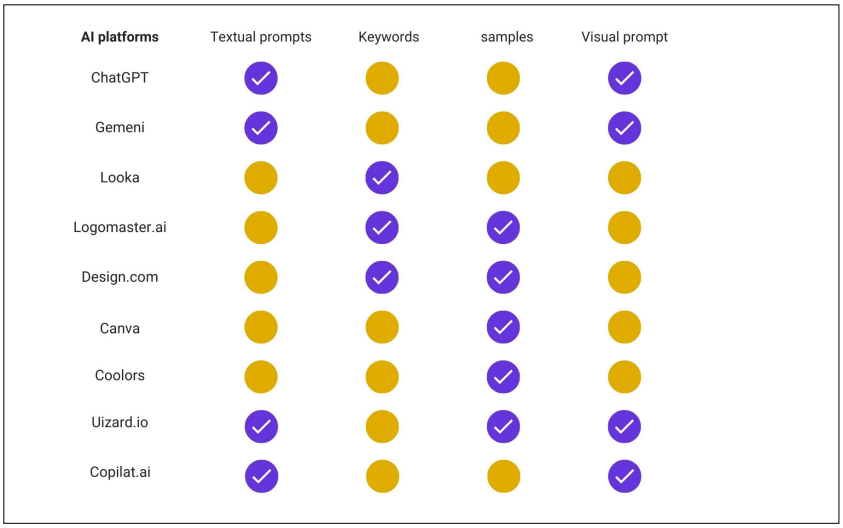


Fig.49
Ai olatform used in deaign process and their different methods of creating results
Graph created by auther 2024

Conclusion

5

5.1 Future of AI in (visual) communication

AI Chatbots are transforming the way businesses communicate with customers. By offering 24/7 support and automated assistance, AI can handle basic inquiries, troubleshoot problems, and even complete simple transactions, all without human intervention. This allows human customer service representatives to focus on more complex issues, increasing efficiency and improving customer satisfaction.[129]

In the field of visual communication, Simply put, AI models can produce stunningly realistic visuals from simple text descriptions. It's a capability that was almost unthinkable just a few short years ago but is now ushering in a new era of creative possibilities for artists, designers and brands alike, for better or worse. [130]

On the other hand, It is important to mention that, While the technical capabilities of AI image generators are intriguing, we all know true creative brilliance comes from humans. Visual communication, particularly when it comes to branding for a business, isn't just about aesthetics, it's storytelling in a way that sparks genuine emotion and connection with the customers.

AI is the shiniest new toy in the creative toolbox, and clearly won't be going away in a hurry. But, like with all toys, the true potential of AI can only be unlocked through human creativity and imagination. The more we play and discover its possibilities, the more we stand to learn from AI. The most impactful designs throughout history have come from daring creative spirit and AI is still just a tool – albeit an extraordinarily powerful one.

In the same way that design developed in the 20th century before and after the advent of computers – it is developing in the same way now, only much faster because technology has advanced a lot. One of the most significant changes that AI has brought to graphic design is the automation of routine tasks. AI can perform tasks such as image resizing, color correction, and even creating layouts with high accuracy. The net result saves time and ensures consistency in design. [131]

AI graphic design is reshaping the industry by enhancing efficiency, providing inspiration, and reducing costs. While it has its limitations and concerns, the future promises an exciting synergy between human creativity and AI assistance. Designers should embrace this technology to streamline workflows and create compelling visual content. AI is not here to replace designers but to elevate their craft to new heights.

129. Sascha H. Funk, (2023), The Future of Communication: How AI is Transforming the Way We Connect

130. JON DIGNAM, (2024), AI VS. THE HUMAN TOUCH: WHO WINS IN DESIGN?,

131. kontra.agency, (2024), AI In Graphic Design: The Impact on Visual Communication

5.2 potential future developments and synergies between AI and startup's communication

The potential future developments and synergies between AI and startups' communication are vast and transformative. As AI chatbots continue to evolve, they will further enhance customer service by providing round-the-clock support, efficiently managing basic inquiries, and automating routine transactions, thereby allowing human representatives to tackle more complex issues. In the realm of visual communication, AI's ability to generate realistic visuals from text descriptions is revolutionizing the creative process for designers and brands. This advancement enables startups to produce high-quality visual content rapidly and consistently. However, the true potential of AI is realized when it complements human creativity and decision-making. By automating mundane tasks such as image resizing and color correction, AI tools free designers to focus on more innovative and strategic aspects of communication. The future promises a harmonious integration where AI enhances human creativity, leading to more compelling and effective communication strategies for startups. Balancing AI's efficiency with human oversight will be crucial to avoid biases and ensure the quality and authenticity of the final output. This synergy will not only streamline workflows but also enable startups to deliver personalized and engaging experiences to their customers, fostering stronger connections and driving business growth.

"The future promises a harmonious integration where AI enhances human creativity, leading to more compelling and effective communication strategies for startups. Balancing AI's efficiency with human oversight will be crucial to avoid biases and ensure the quality and authenticity of the final output."

5.3
Key Discoveries

In summary, this study provides a comprehensive exploration of the dynamic relationship between artificial intelligence (AI) and the establishment of startups, particularly in the field of communication. By examining various phases, from the definition and types of AI to the definition of startups and the role of communication within them, we gained a holistic view of the thesis's subject matter.

In the second chapter, we discussed the growing influence of AI at different stages of startup establishment and provided an overview of AI's role not only in communication but also in strategic communication. This offered a clear understanding of how AI can be integrated into business creation and how it can enhance communication processes.

Additionally, the challenges of incorporating AI into startups were studied and analyzed. Transitioning from the theoretical framework to practical application, the final project involved creating a mobile application for an on-demand beauty service in Italy. The aim was to develop the visual communication strategy for this startup using AI tools. This project revealed two main perspectives: AI as a collaborative tool requiring human decision-making for the final output, and AI as an autonomous tool capable of automatically generating communicative results for a brand.

In this project, AI played a crucial role in visual consumer communication. Utilizing large language model (LLM) chatbots like GPT-3 and Gemini, we generated texts and suggestions that assisted in all phases of business creation. Furthermore, AI-based platforms such as Coolors, Canva, Looka, Logo Maker, and Copilot were employed to create visual elements for the brand, including the color palette, logo, and illustrations.

This trial confirmed that AI significantly impacts communication, underscoring its potential to transform how startups interact with consumers visually.

But Overreliance and decision bias is always obvious . Blindly relying on AI systems for decision making without proper human oversight can lead to overreliance and potential biases. The right balance must be struck between human judgment and AI assistance to avoid undue reliance or abdication of responsibility. [132]

Thus, it is crucial to point out that the final results created by AI in the final project were not always satisfactory and related based on the designer's attitude and also the survey that had been engaged asking visual communication designers. In this survey that had been done with the participation of 15 designers, I first provided them all the reults that had been cretaed using AI, in different stages, and then I asked them their opinion on if the results are satisfactory, related and uable based on their Idea or no, the survey can bee seen

132. The Upwork Team (2024), How AI Is Used in Decision-Making

in figure 50.

	yes	Not sure	No
1. Would you use the suggestions made by **CHATGPT 3,** for the Naming of the platform? if yes, which name would you choose?			
2. Would you use the logos created by **Looka**? if yes, which logo would you choose?			
3. Would you use the logos created by **logomaster.ai**? if yes, which logo would you choose?			
4. Would you use the logos created by **design.com**? if yes, which logo would you choose?			
5. Would you use **canva** to create color palettes for your brand and platform?			
6. Would you use **coolors** to create color palettes for your brand and platform? which one you prefer to use mostly? Canva or coolors?			
7. Would you use **copilot ai** to create illustrations for your brand and platform? if yes, which illustrations would you choose mostly?			
8. Would you ask **gemini** to guide you on creating and designing your platform's mobile application ? if yes, which suggestions would you use mostly?			
9. Would you use the results generated by **Uizard.io** for Ui design ? if yes, which pages would you use?			

Fig.50
Survey asked 15 designers for the results created by AI in design process of final project
Survey created by auther
2024

As shown in Figure 51, the final survey results indicate that AI-generated outcomes were not consistently satisfactory or usable. Consequently, in the final project, AI was utilized exclusively for the naming process. For color palette design, AI was employed to an extent of 85%, for UI mobile application design 35%, for logo design 20%, and for illustrations only 10%, This means that in other stages human desicion had to be fully involved to get the desirable result.

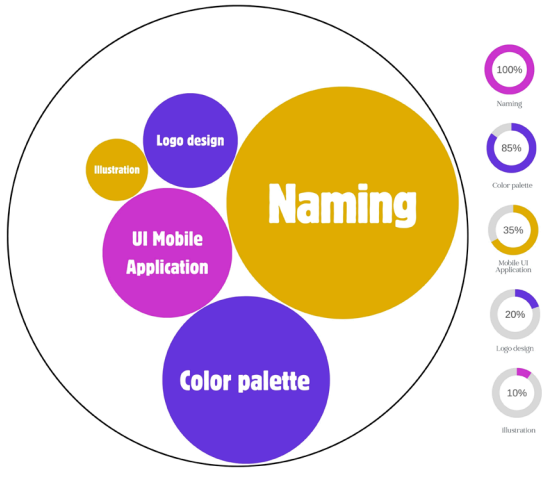


Fig.50
The persantage showing how much AI was helpful in each stage of visual communicaion design
Survey created by auther
2024

Finally it is important to note that, To address the diverse needs across different locations in Italy and to cater to the varying requirements of customers based on their city, AI-driven personalization tools were utilized. These tools enable the application to tailor suggested services for consumers based on their recent searches on the platform. Thus, when a consumer accesses the application, they receive service recommendations specifically aligned with their location and search history.

Actually, AI-driven personalization is the use of artificial intelligence and machine learning algorithms to analyze data and predict the preferences and behaviors of individual users. By leveraging AI, companies can create hyper-personalized experiences for their customers across various touchpoints, such as websites, mobile apps, email campaigns, and more. Enhancing customer Engagement: Improving Conversions By offering personalized experiences, companies can drive higher conversion rates and increase sales, generating Better Customer Insights with AI-powered algorithms can provide valuable insights into customer behavior, preferences, and trends, helping companies make more informed decision and Increasing Customer Loyalty was some of the aims of integrating Ai for the personalization of the mobile application designed.

It is a fact that AI now may save tons of time, energy and consequently money especially when it comes to creating a startup.but it is critical to mention that although AI at times can make very satisfying results, mostly human decision is needed to elaborate them and also decide on which result can be used that was generated by AI.

Looking forward, the potential of AI in visual communication is boundless. As AI technologies continue to advance, their ability to assist and augment human creativity will only grow stronger. The future of AI in this field will see even more sophisticated tools that not only automate routine tasks but also provide deeper insights and innovative solutions that were previously unimaginable. Startups will benefit from faster, more efficient, and highly personalized communication strategies that resonate more deeply with their target audiences. By leveraging AI, businesses can create more engaging, authentic, and emotionally compelling visual content, thereby forging stronger connections with their customers. The true power of AI lies in its synergy with human ingenuity, and as this relationship evolves, it will redefine the landscape of visual communication, setting new standards for creativity and effectiveness in the startup ecosystem.

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